

Flow Matching Predictive Control with Constraints

Master's Thesis in Informatics · Defence

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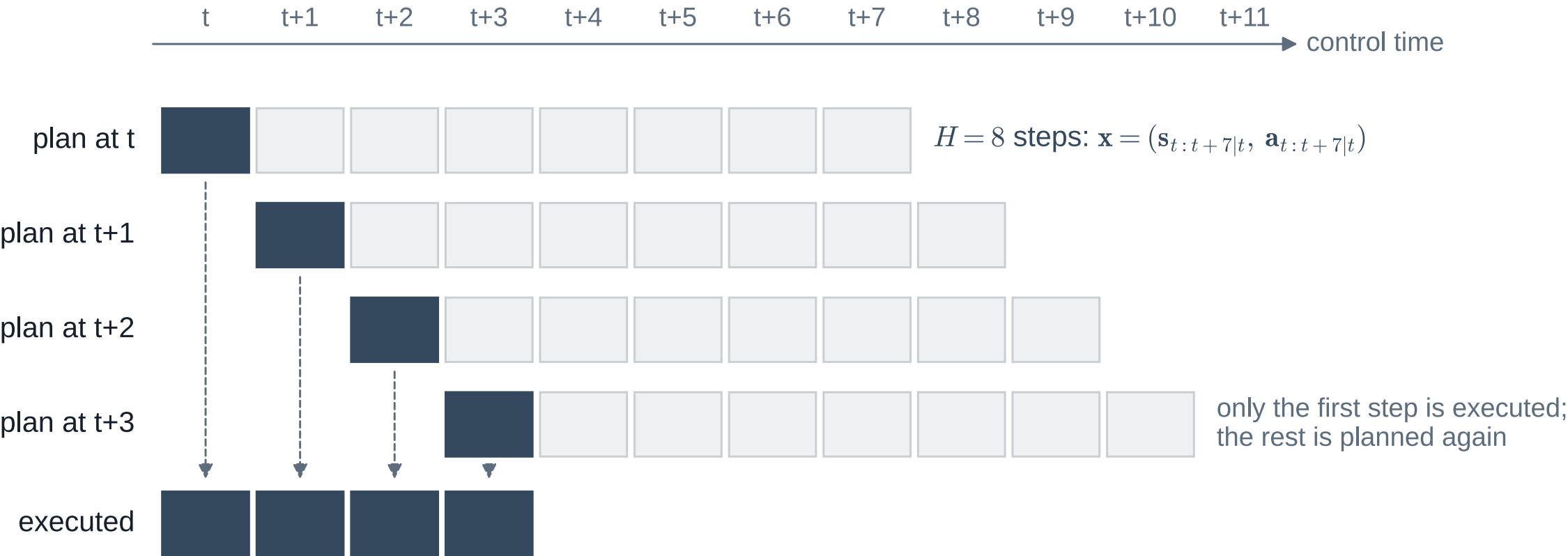
DPCC: projection inside a fixed denoising sequence

- Diffusion planners capture multimodal demonstrations, and Diffusion Predictive Control with Constraints (DPCC) projects their denoising iterates onto the feasible set.
- In the inherited implementation the denoising sequence is fixed at training and repeated at every control step.
- Every control step therefore pays for the whole sequence, and the projection with it.

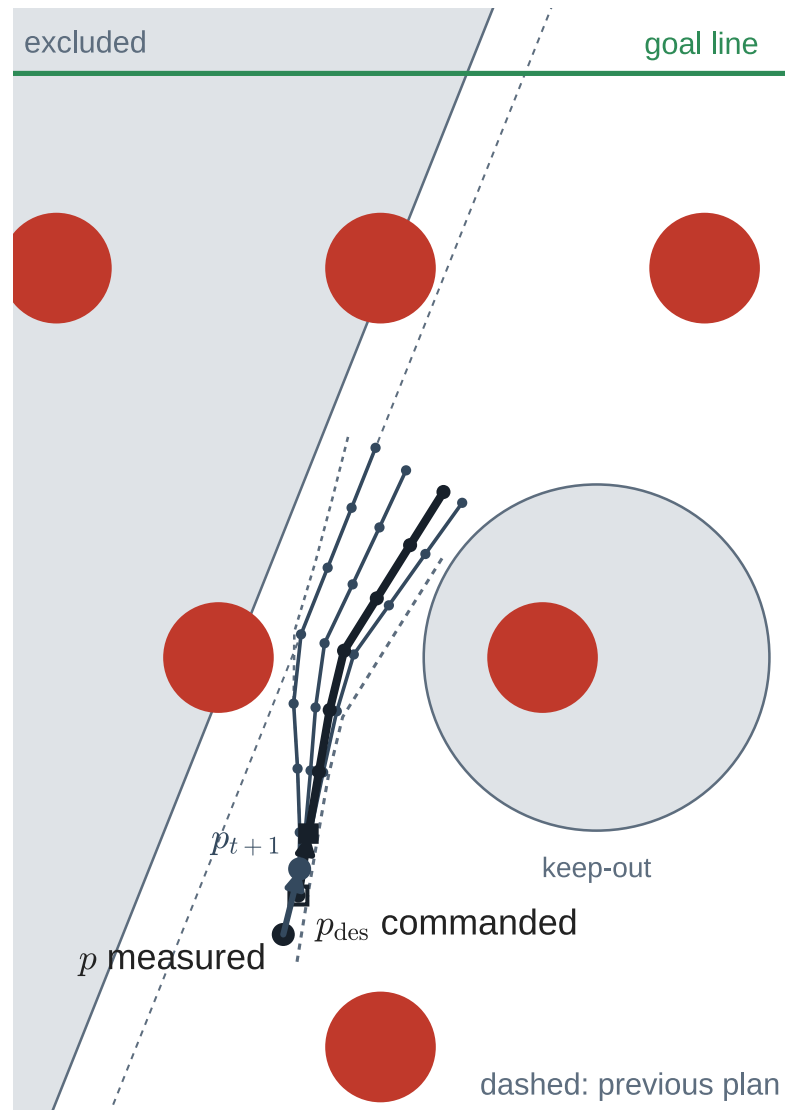
This thesis replaces diffusion with deterministic transport along an ordinary differential equation, whose step budget is chosen at deployment.

The question: can flow-based models reach the baseline's success within the constraints with fewer sampling steps and less planning time?

Receding horizon: plan eight steps, execute the first, plan again



One control step, block by block, on D3IL-avoiding



1 · Observation — the commanded and the measured position, $\mathbf{o}_t = (p_{\text{des},t}, p_t)$.

2 · Candidates — $B = 4$ plans, each drawn from independent noise.

3 · Sampling with projection — the model's K steps; the projection constrains the iterates or the predicted endpoint.

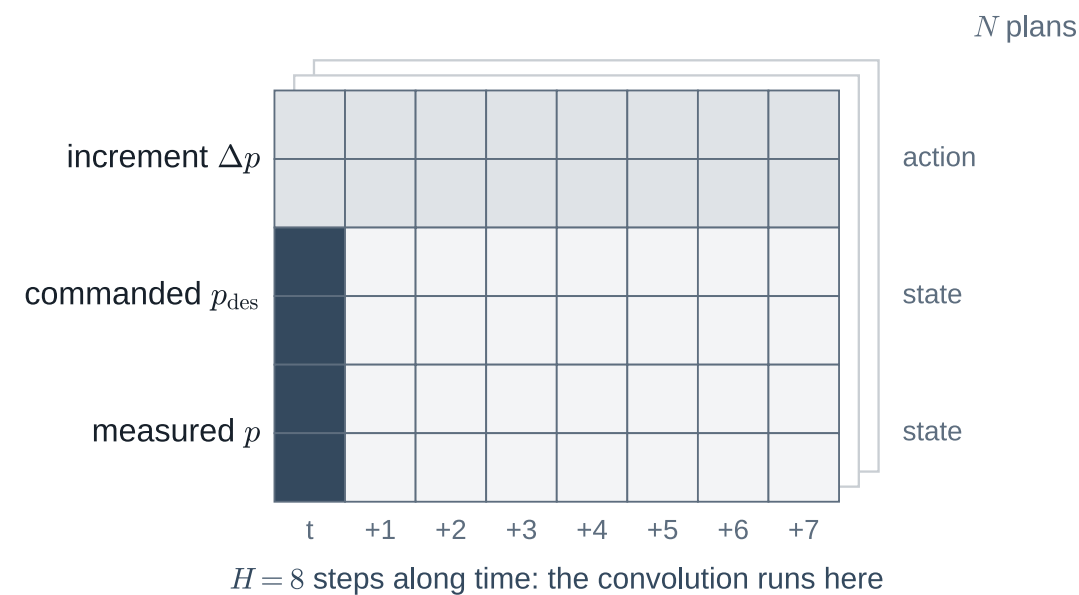
4 · Candidate selection — temporal consistency: the candidate closest to the previous plan.

5 · First increment → setpoint — $p_{\text{des},t+1} = p_{\text{des},t} + \Delta p_t$.

6 · Tracking controller — differential IK → joint PD moves the arm towards the setpoint; the next observation starts the next step.

The thesis's scene (obstacles, goal line, the top-right-hard constraint set); the plans are drawn for illustration.

The network reads a plan as an $8 \times d$ signal



The plan tensor of D3IL-avoiding, $(N, 8, 6)$. Dark: the known current state, imposed by inpainting and masked in the loss.

Environment	Plan in / out	State anchor
D3IL-avoiding	$(N, 8, 6)$	$(N, 4)$
D3IL-aligning	$(N, 8, 9)$	$(N, 6)$
UAV-corridor, UAV-s-curve	$(N, 8, 12)$	$(N, 9)$

D3IL-aligning adds two RGB views, $2 \times (N, 1, 3, 96, 96)$, encoded to $(N, 128)$. UAV-pillars flies the D3IL-avoiding models.

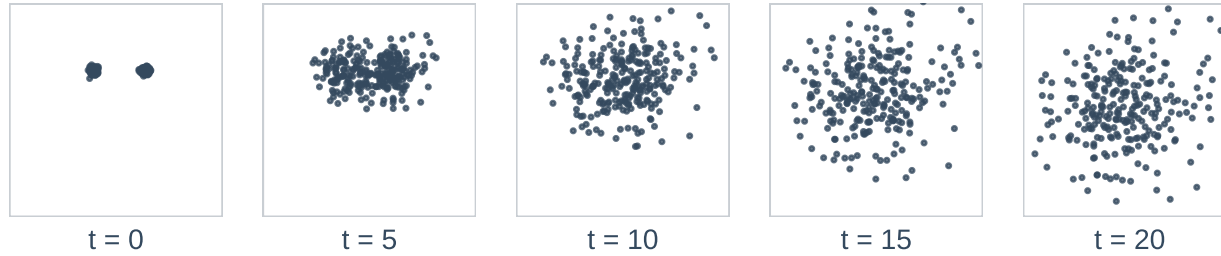
- A temporal U-Net: convolution along time, channel widths $(d, w, 2w, 4w, 8w)$, three stride-two stages ($8 \rightarrow 4 \rightarrow 2 \rightarrow 1$) with skip connections.
- Conditioning is a per-channel bias in every block: the time embedding of τ , plus h for Analytic-MeanFM and CI-MeanFM, plus the image latent on D3IL-aligning.
- The output has the plan's shape: v_θ , u_θ , or the baseline's noise ϵ_θ .

Diffusion: predict the noise, remove it step by step

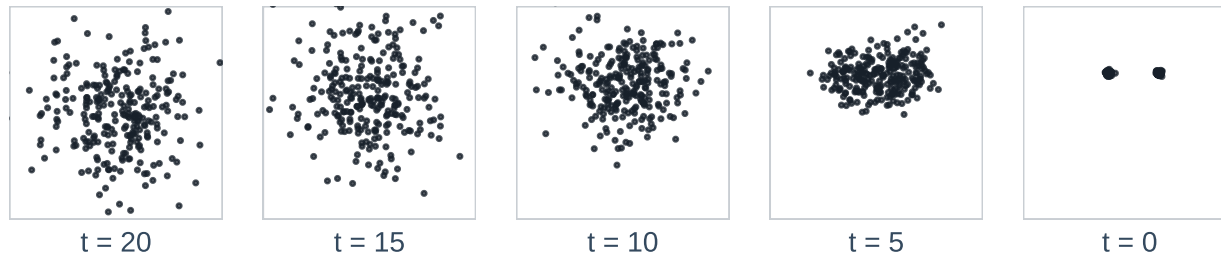
forward (training): add noise step by step



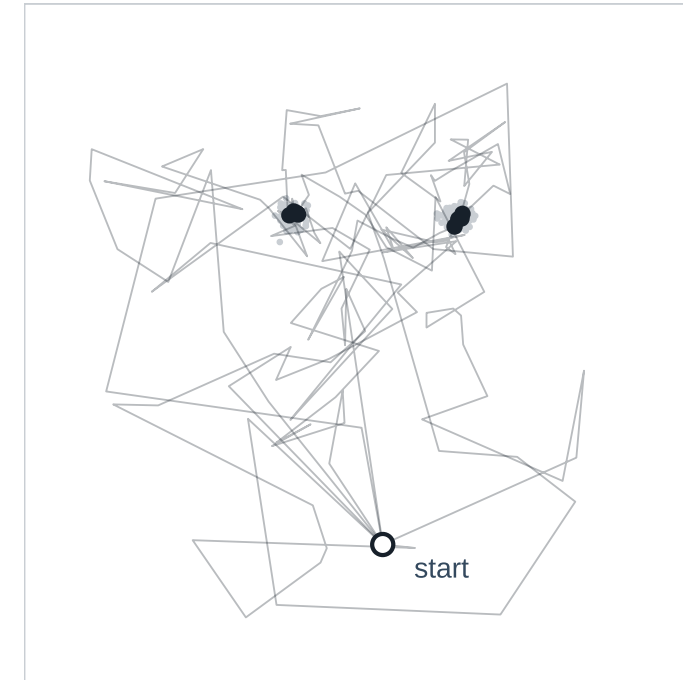
data: two modes



reverse (sampling): the network removes the noise, $K = 20$ steps



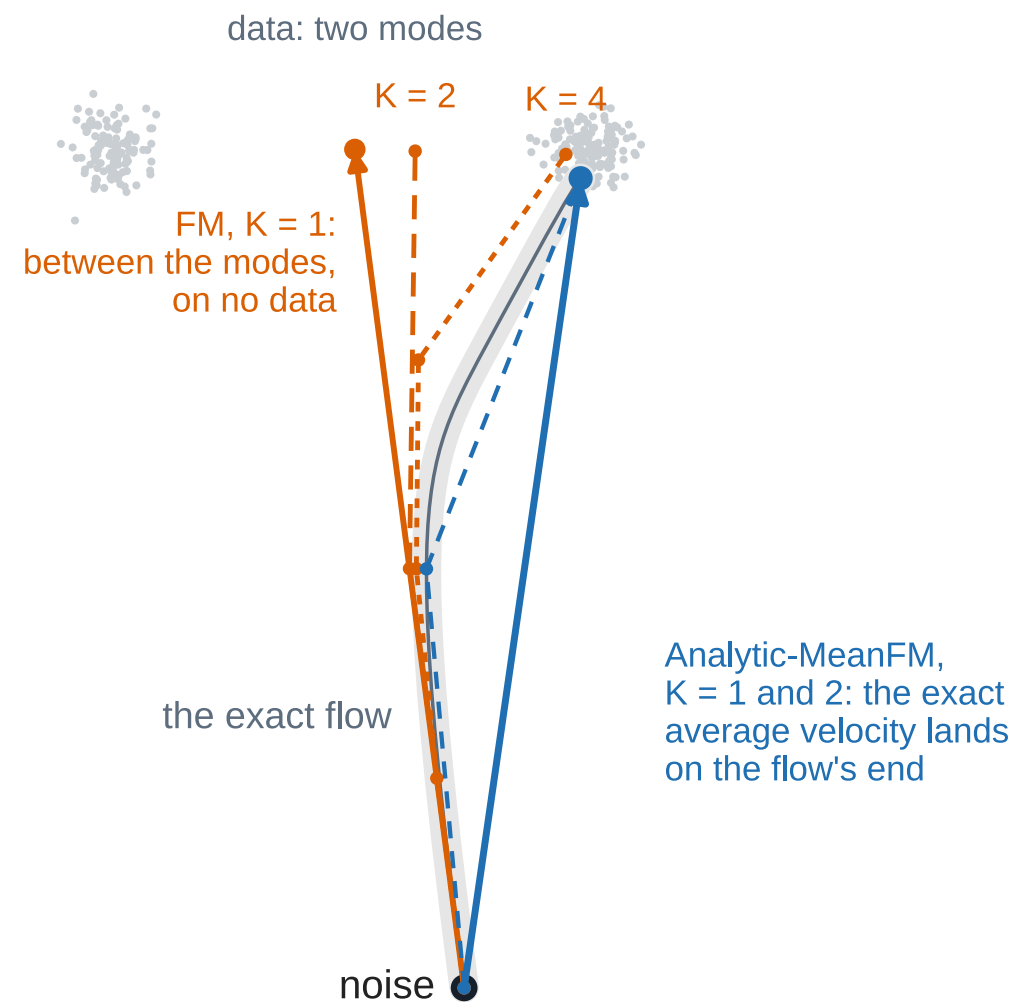
one start, sampled 8 times



- A network is trained to predict the noise added at each step; generation starts from pure noise and removes it step by step.
- The number of steps is tied to the training schedule: fixed when training ends ($K = 20$ in DPCC).
- The sampler adds noise at every step but the last: plans for the same observation can differ.

Toy: a two-mode Gaussian mixture, the exact noise prediction, a cosine schedule; computed, not thesis data.

Flow matching: a velocity field transports noise to data



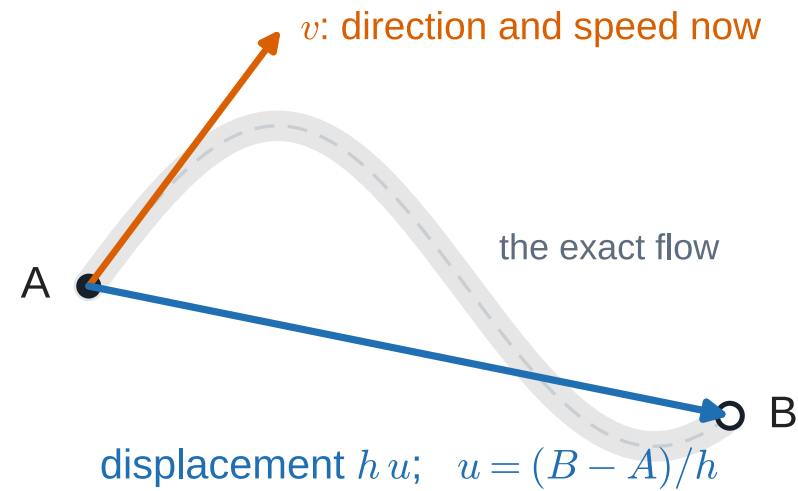
	learns	steps K	sampling
Diffusion	the noise	fixed at training	adds noise
FM	velocity v	chosen at deployment	deterministic
Analytic-MeanFM, CI-MeanFM	average velocity u	chosen at deployment	deterministic

- Straight per-sample paths average into a curved field, and coarse steps along a curved field are inaccurate.
- One evaluation of an accurate average velocity covers the whole interval.

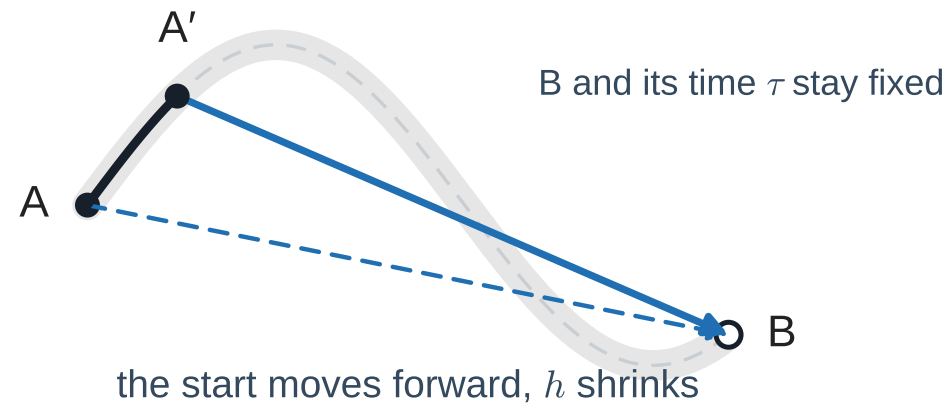
The same toy; the exact flow is solved finely. Computed, not thesis data.

A car on a winding road: what the average-velocity objectives learn

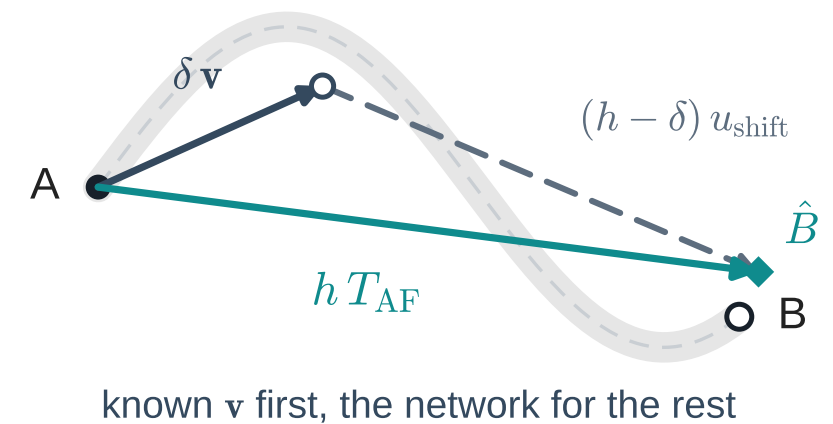
1 · velocity and average velocity



2 · Analytic-MeanFM: keep the destination



3 · CI-MeanFM: a two-part target



$$h u(\mathbf{x}_r, r, h) = \mathbf{x}_{r+h} - \mathbf{x}_r$$

one step from the start:

$$\hat{\mathbf{x}}_1 = \mathbf{x}_0 + u_\theta(\mathbf{x}_0, 0, 1)$$

$$u = v + h \frac{d}{dr} u$$

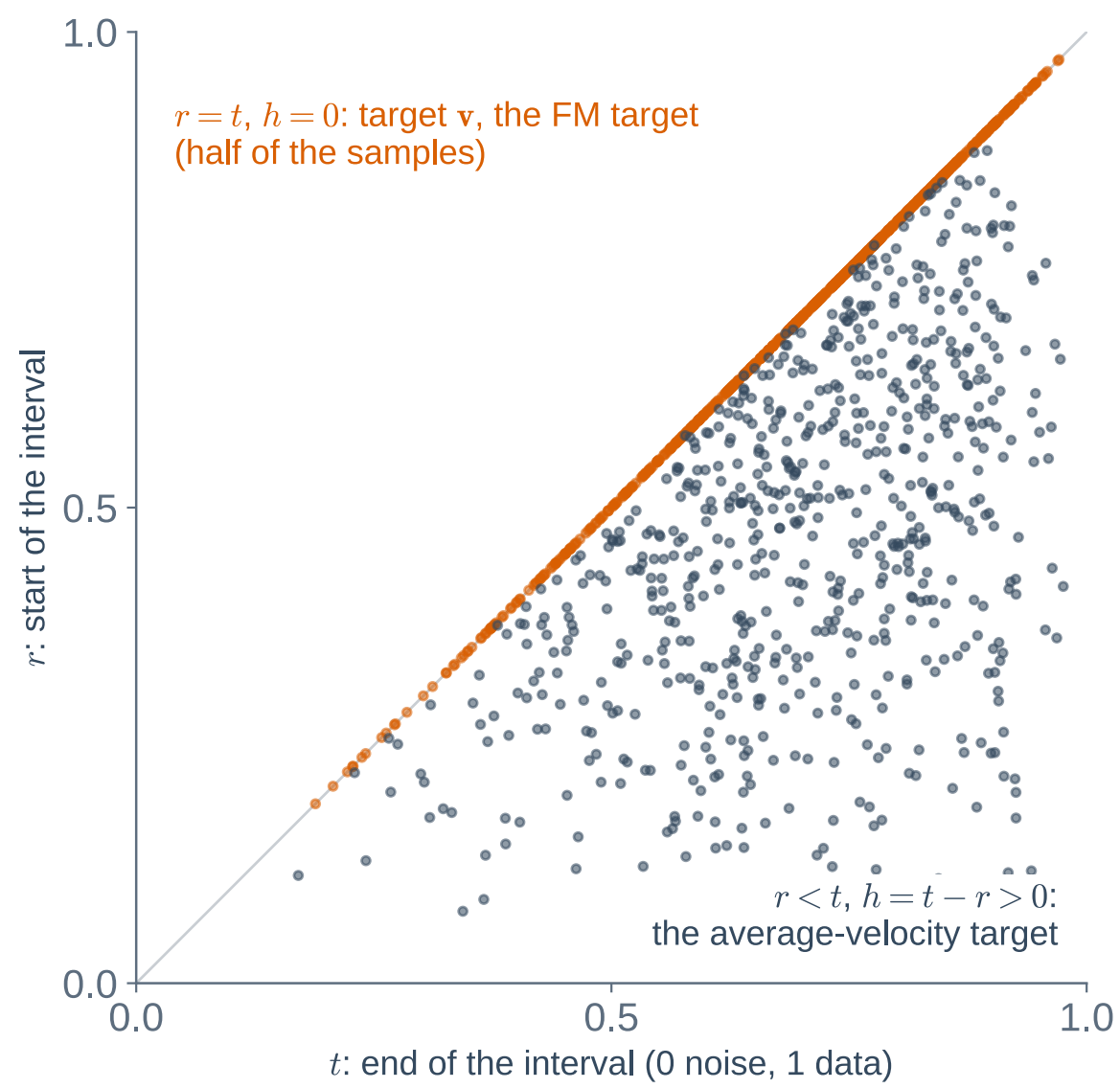
target $\text{sg}(v + h \text{JVP})$: a derivative at the current input

$$h u_{\text{tgt}} = \delta v + (h - \delta) u_{\text{shift}}$$

$\delta = \alpha h$; a prediction at a shifted input; α from 1, held at 0.2 in training

The road is the exact flow: a trajectory through the space of candidate plans; h is generation time, not seconds. A drawing, not thesis data.

Training: one average-velocity network, half of its samples anchored at $h = 0$



sample	target of the average head u_θ
$h = 0$, half of the samples	$\boldsymbol{v} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, the FM target
$h > 0$, Analytic-MeanFM	$\text{sg}[\boldsymbol{v} + h \text{JVP}]$
$h > 0$, CI-MeanFM	$\alpha \boldsymbol{v} + (1 - \alpha) \text{sg}(u_\theta(\cdot))$, weighted by α

- A second, auxiliary head v_θ regresses $\boldsymbol{v}$ on every sample; sampling uses u_θ only.
- The two objectives share the network, its auxiliary head, the zero-interval samples and the sampler; they differ in how the average-velocity target is built, in its loss weight and in the curriculum (α from 1, held at 0.2).

Three objectives inside the same constrained controller

FM

instantaneous-velocity matching

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left\| v_{\theta}((1 - \tau)\mathbf{x}_0 + \tau\mathbf{x}_1, \tau, \mathbf{c}) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|_2^2$$

Analytic-MeanFM

analytic average-velocity matching

$$u_{\text{tgt}} = \text{sg} \left[\mathbf{v} + h \text{JVP} \left(u_{\theta}; (\mathbf{x}_r, r, h), (\mathbf{v}, +1, -1) \right) \right]$$

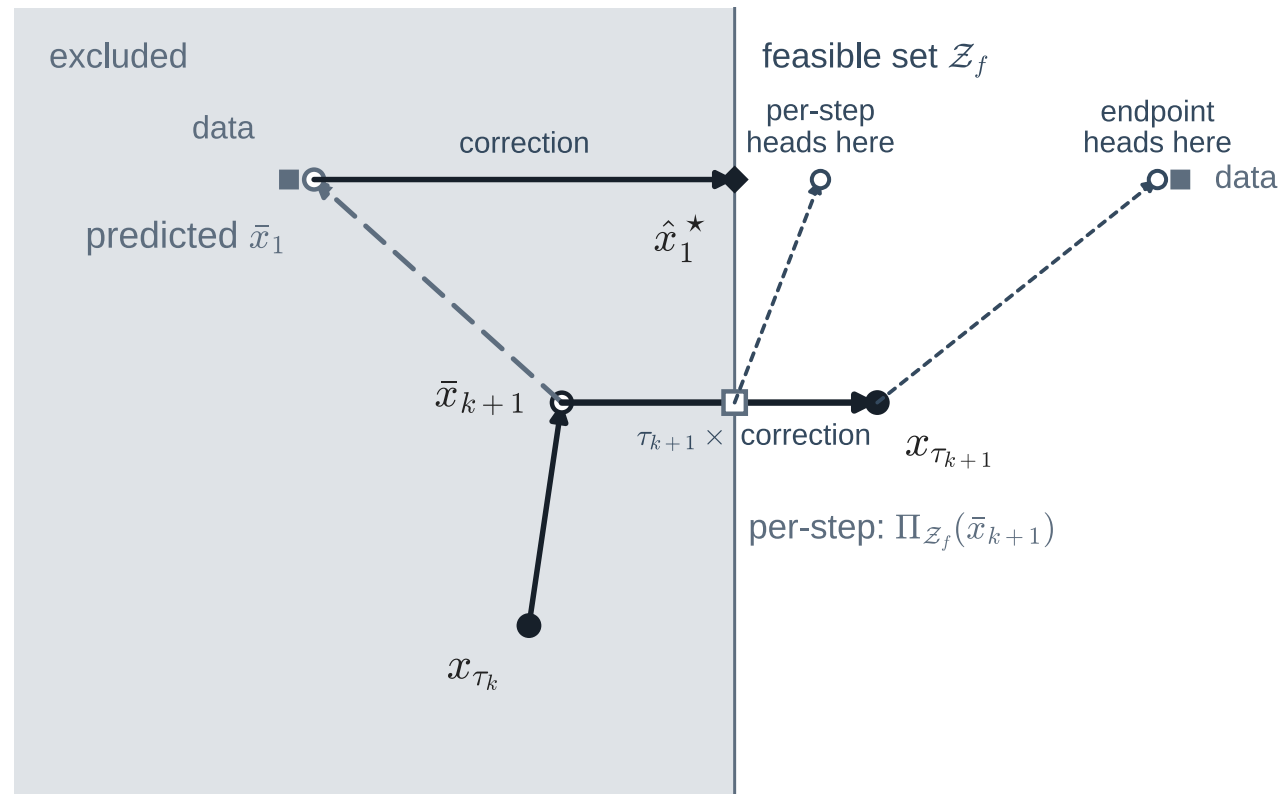
CI-MeanFM

consistency-interpolated average-velocity
matching

$$u_{\text{tgt}} = \alpha \mathbf{v} + (1 - \alpha) \text{sg} \left(u_{\theta}(\mathbf{x}_r + \delta \mathbf{v}, r + \delta, h - \delta) \right), \quad \delta = \alpha h$$

$\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$; sg: stop-gradient. Backbone, projection, data and evaluation protocol are kept; the number of network evaluations becomes a choice made at deployment. Both average-velocity objectives also train on zero-interval samples and an auxiliary instantaneous head.

Per-step projection constrains the current iterate, endpoint projection a predicted final plan



Toy: two data points, a halfspace, one step at $\tau_k = 0.5$, $K = 4$; computed, not thesis data.

Each point stands for a whole plan: the squares are two demonstrated plans, the white half is the feasible set $\mathcal{Z}_f$. Projecting moves a point to the closest feasible point.

→ 1. Step the sampler once, unguided

$$\bar{\mathbf{x}}_{k+1} = \mathbf{x}_{\tau_k} + \Delta\tau v_\theta(\mathbf{x}_{\tau_k}, \tau_k, \mathbf{c})$$

----> 2 · **Predict** the final point: here, the excluded data point

$$\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_{k+1} + (1 - \tau_{k+1}) v_\theta(\bar{\mathbf{x}}_{k+1}, \tau_{k+1}, \mathbf{c})$$

→◆ **3 · Project** the prediction to the closest feasible point

$$\hat{\mathbf{x}}_1^\star = \arg \min_{\hat{\mathbf{x}}_1 \in \mathcal{Z}_f} \|\hat{\mathbf{x}}_1 - \bar{\mathbf{x}}_1\|_2^2$$

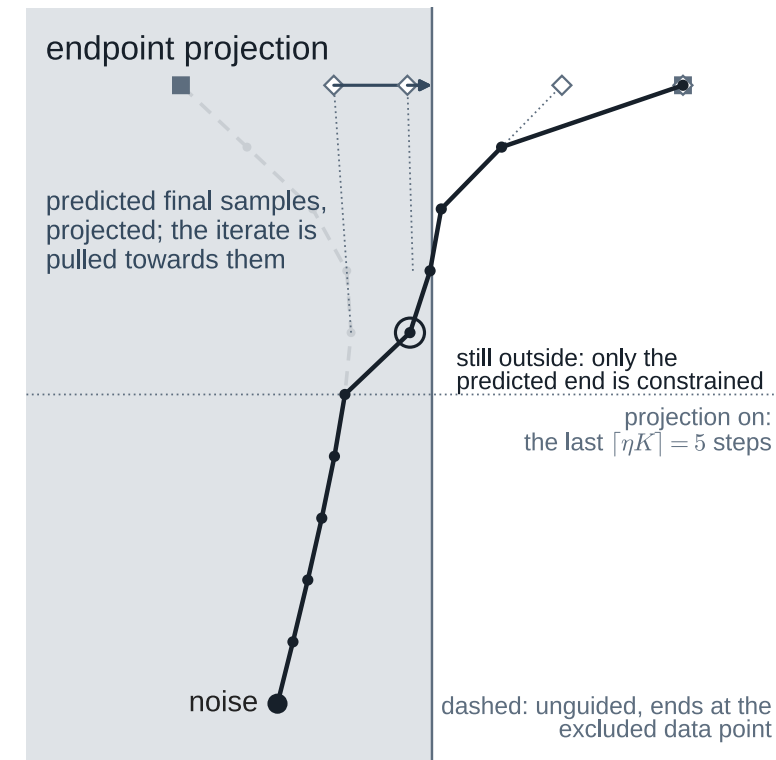
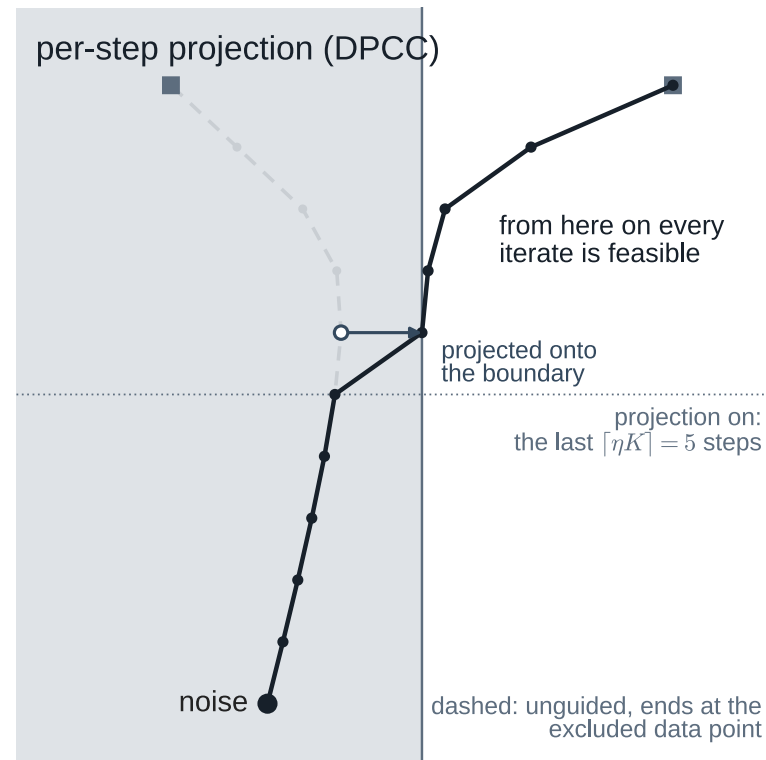
→● **4 · Pull back:** move the iterate by τ_{k+1} times the same correction

$$\boldsymbol{x}_{\tau_{k+1}} = \bar{\boldsymbol{x}}_{k+1} + \tau_{k+1}(\hat{\boldsymbol{x}}_1^* - \bar{\boldsymbol{x}}_1)$$

- **Per-step (DPCC)** projects the iterate itself: $\mathbf{x}_{\tau_{k+1}} = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1})$.

.....►◦ **Where each now heads:** per-step, between the two data points; endpoint, to the feasible one.

One sampling run: per-step against endpoint projection



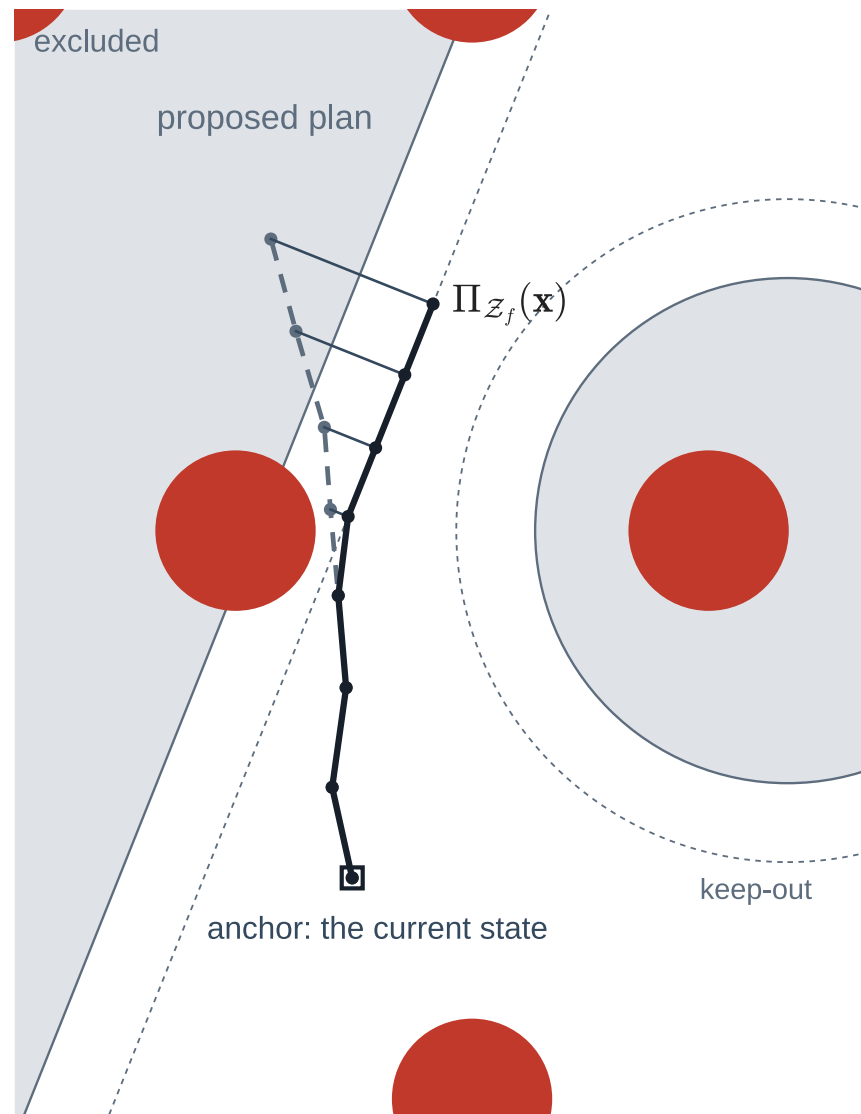
- **Per-step (DPCC)** corrects the current iterate. From the threshold on, every iterate is projected and therefore feasible; here only the first projection moves it.

- **Endpoint** corrects the plan the sampler is heading towards. The iterate follows, damped by τ , and can itself still be outside. $\text{NFE} = K + n_{\text{guide}} = 10 + 4$.

At the last step both project the finished sample: both end at the feasible data point, the unguided run at the excluded one.

Toy: two data points and a halfspace, $K = 10$, $\eta = 0.5$; computed, not thesis data.

The projection: the closest plan that satisfies the constraints



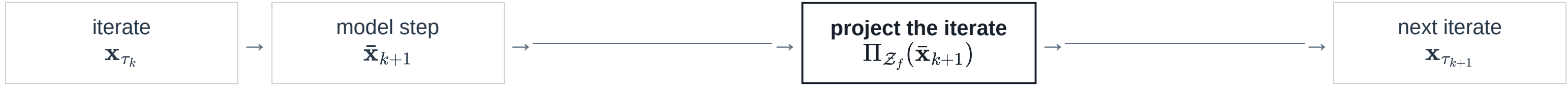
$$\Pi_{Z_f}(\mathbf{x}) = \arg \min_{\mathbf{z}} \frac{1}{2} \|\mathbf{z} - \mathbf{x}\|^2$$

- subject to the **dynamics rows** of the first-order model, the **bounds and halfspaces**, and one **quadratic keep-out inequality** per horizon step and obstacle. The first block is fixed at the measured state.
- The constraints are **tightened** by a margin for the model mismatch.
- **Goal-independent**: its objective is proximity to the proposed plan. Success is not its objective.
- A non-linear program, since the keep-out rows are non-convex, solved by SLSQP. Its cost $\frac{1}{2} \|\Pi_{Z_f}(\mathbf{x}) - \mathbf{x}\|^2$ is what the cumulative-cost selection rule sums.

The thesis's scene (top-right-hand, tightened), drawn pointwise; the dynamics rows are left out of the picture.

Two ways to put the projection into sampling

Per-step projection (DPCC): project the current iterate



Guidance. A feasibility likelihood around $\mathcal{Z}_f$, expanded to first order, gives a step to the projection; in the diffusion sampler $\mathbf{x}^{k-1} = \Pi_{\mathcal{Z}_f}(\boldsymbol{\mu}_\theta + \tilde{\sigma}_k \boldsymbol{\epsilon}^k)$. Early in sampling the iterate is still far from the data, so the keep-out rows are evaluated on positions the sampler has not yet produced.

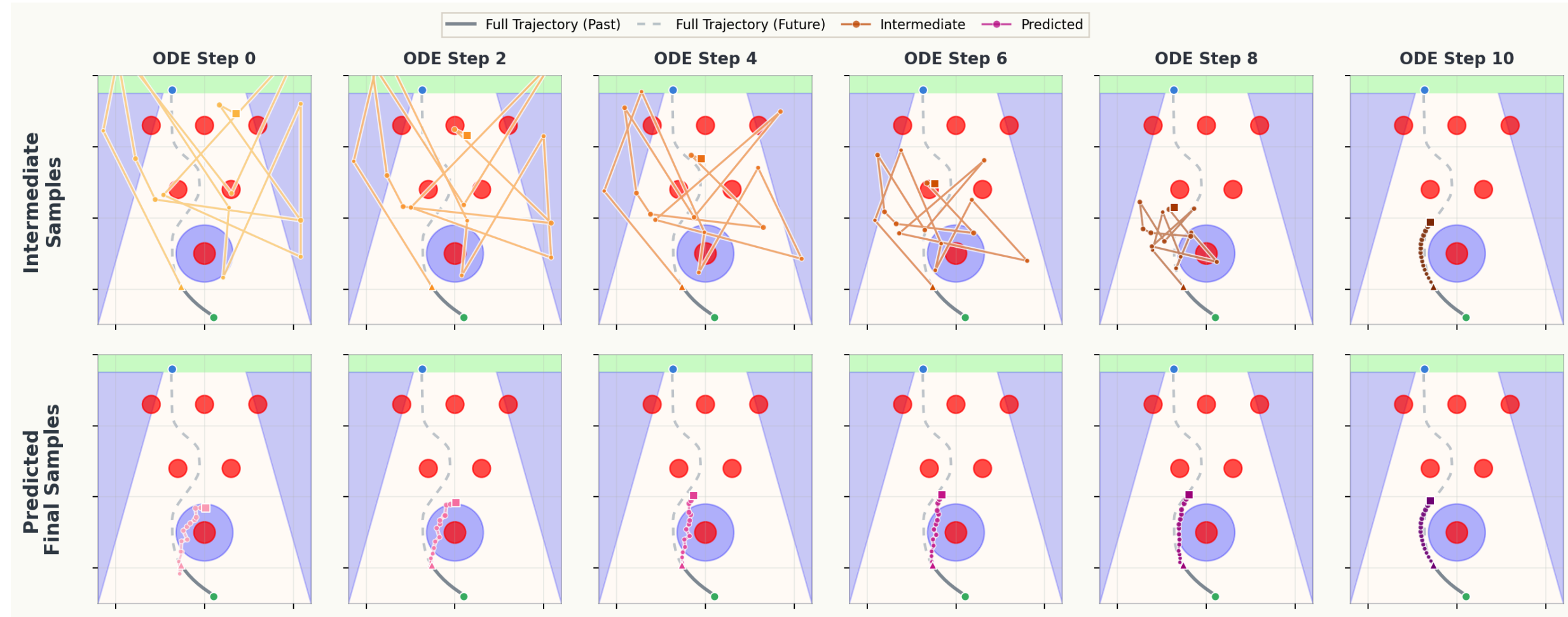
Endpoint projection (HardFlow): project the predicted final plan



Optimal control. Steer the learned flow with the least control effort so that only the final sample is feasible: $\min_{\mathbf{u}} \lambda \int_0^1 \frac{1}{2} \|\mathbf{u}_\tau\|^2 d\tau$ s.t. $\dot{\mathbf{x}}_\tau = v_\theta + \mathbf{u}_\tau$, $\mathbf{x}_1 \in \mathcal{Z}_f$. Solved one sampling step at a time, on the predicted final plan, so that the constraint set is not distorted by the network.

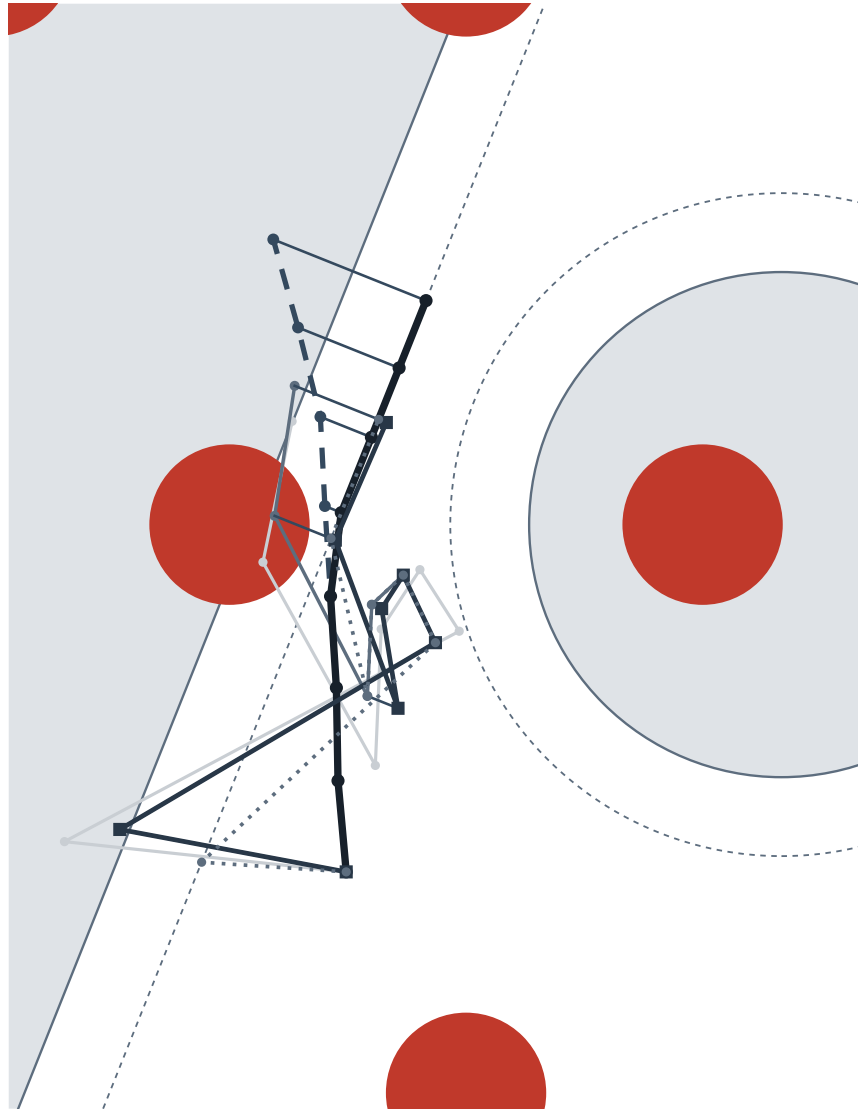
Both solve the same program on the same feasible set with the same solver, from a threshold step to the end of sampling. The terminal cost is zero here, so the HardFlow solve is the projection of the predicted plan.

Endpoint projection constrains a trajectory from the first step on



- **Upper row:** the sample the solver holds, far from a trajectory until the last steps. Per-step projection poses its program on such an iterate.
- **Lower row:** its predicted final sample, a trajectory from the first step on: the point endpoint projection constrains.

Endpoint projection, one guided step, on a plan



The first guided step, $\tau_k = 0.5 \rightarrow \tau_{k+1} = 0.6$ of $K = 10$. The iterate is still noisy.

—•— **1 · Step** the sampler once, unguided

$$\bar{\mathbf{x}}_{k+1} = \mathbf{x}_{\tau_k} + \Delta\tau v_{\theta}(\mathbf{x}_{\tau_k}, \tau_k, \mathbf{c})$$

----- **2 · Predict** the final plan, in closed form

$$\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_{k+1} + (1 - \tau_{k+1}) v_{\theta}(\bar{\mathbf{x}}_{k+1}, \tau_{k+1}, \mathbf{c})$$

—•— **3 · Project** the predicted plan

$$\hat{\mathbf{x}}_1^* = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_1)$$

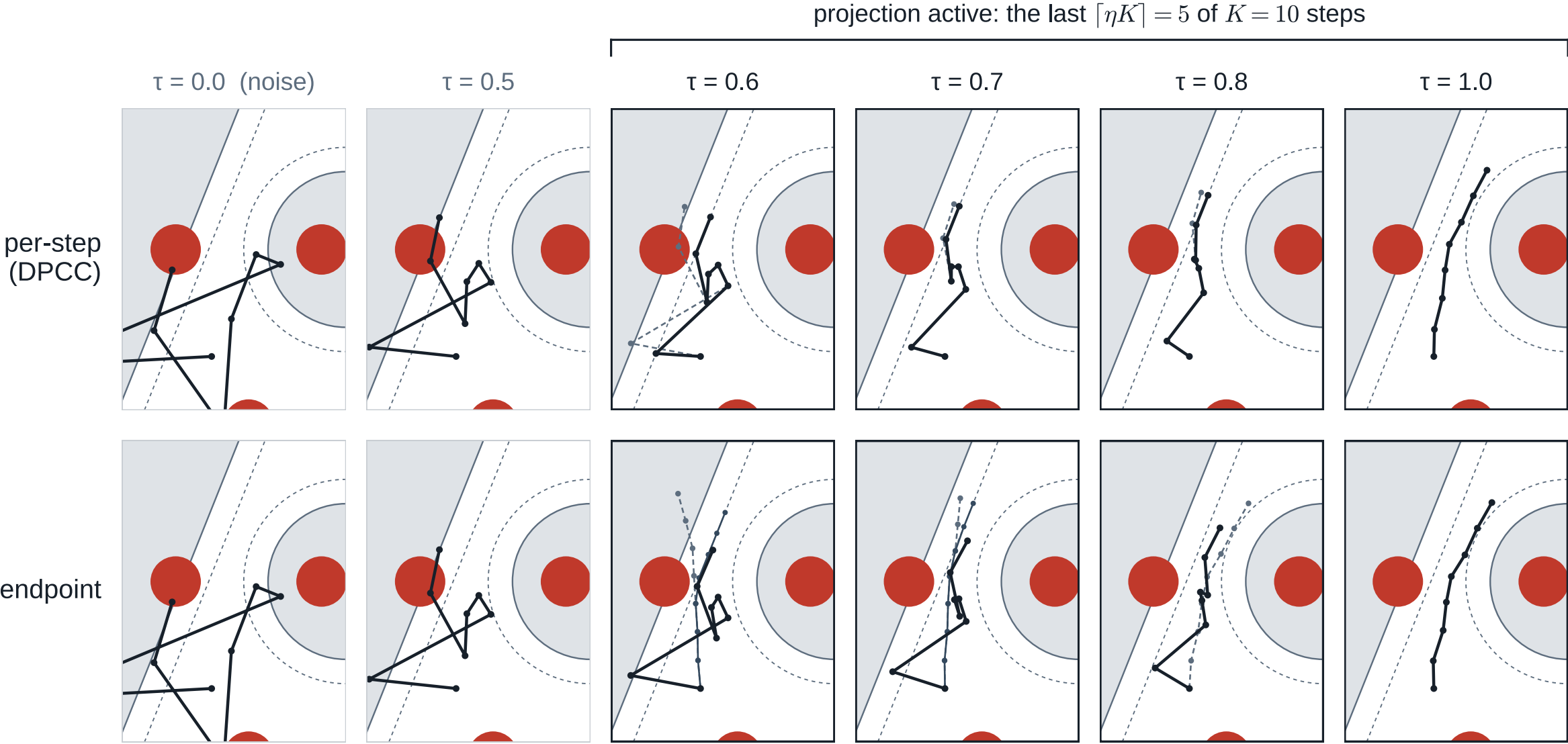
—■— **4 · Pull back** the correction, damped by τ_{k+1}

$$\mathbf{x}_{\tau_{k+1}} = \bar{\mathbf{x}}_{k+1} + \tau_{k+1} (\hat{\mathbf{x}}_1^* - \bar{\mathbf{x}}_1)$$

..... Per-step projection would project the noisy iterate itself, $\Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1})$.

The plan-space toy on the thesis's scene. Computed, not thesis data.

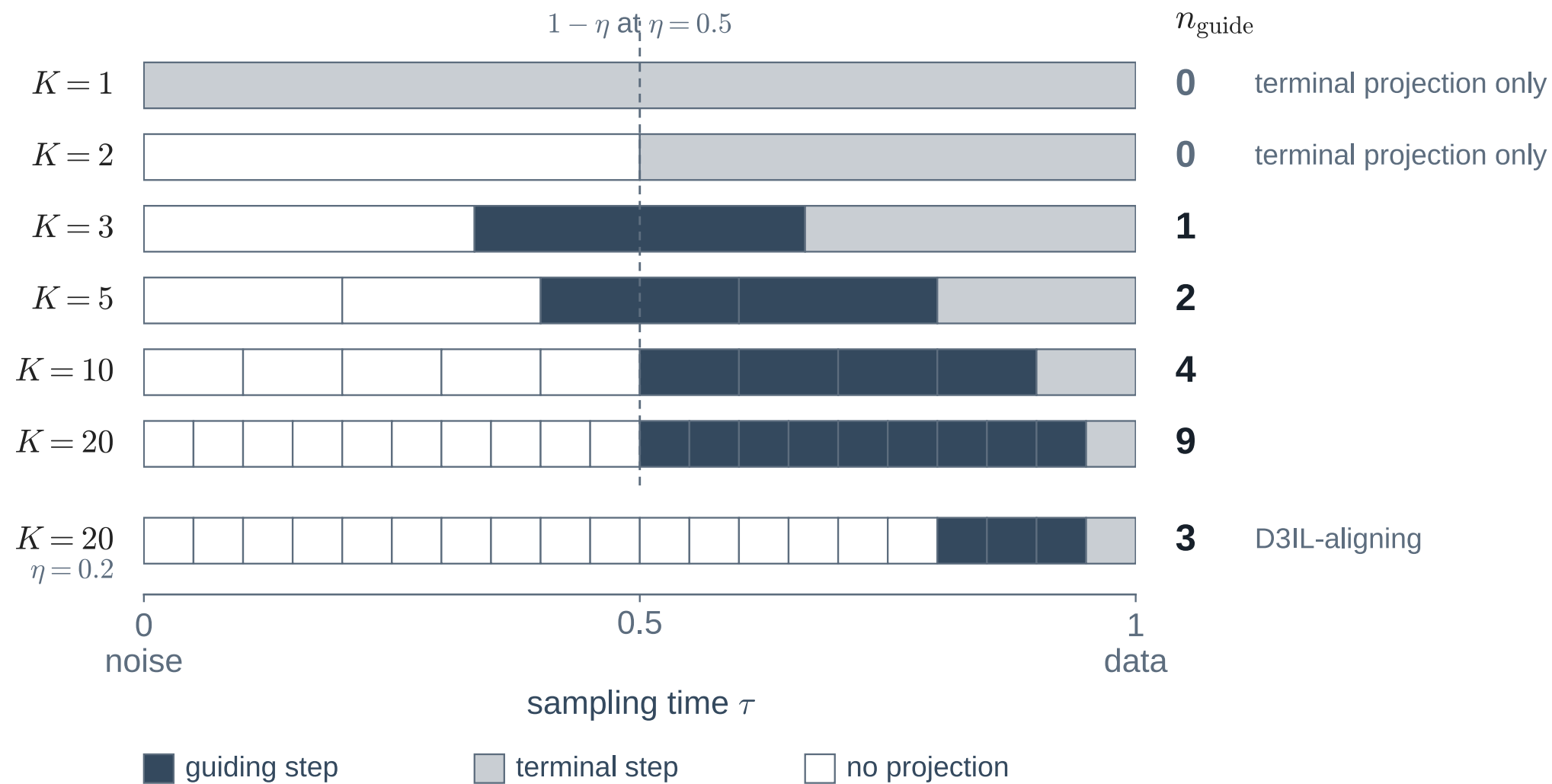
One sampling run, step by step: per-step against endpoint projection



Per-step: the Euler step (dashed) is projected (solid). **Endpoint:** the predicted plan (dashed) is projected (thin), and the iterate (solid) follows.

The plan-space toy on the thesis's scene: two demonstrated modes, the exact flow-matching field, one noise seed. Computed, not thesis data.

Endpoint projection guides only where the budget leaves a step before the last



$n_{\text{guide}}(K, \eta) = \lceil \eta K \rceil - 1$ for $\eta > 0$: the active steps are the last $\lceil \eta K \rceil$, and the terminal one does not guide.

Research questions

- **RQ1 — Generative model.** Can the flow-based models reach the diffusion baseline's task success and constraint satisfaction with fewer sampling steps and less planning time per control step?
- **RQ2 — Constraints.** Does endpoint projection reach the goal with fewer constraint violations, fewer control steps or less computation than the per-step projection of DPCC?
- **RQ3 — Transfer.** Do the answers to RQ1 and RQ2 hold when the observation becomes camera images, when the task requires aligning an object through contact, and when the platform is a quadrotor?

Three steps: **D3IL-avoiding**, the benchmark of DPCC unchanged → **D3IL-aligning**, observed through cameras → **quadrotor**, three scenes.

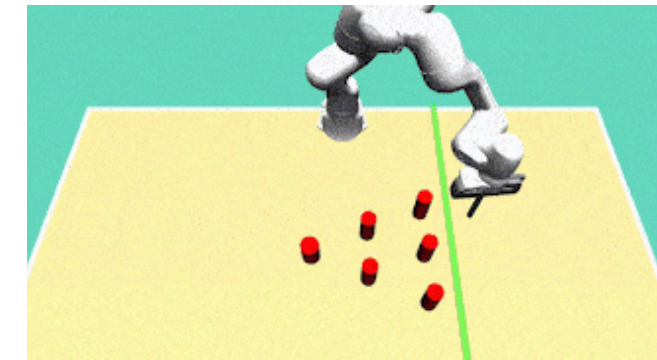
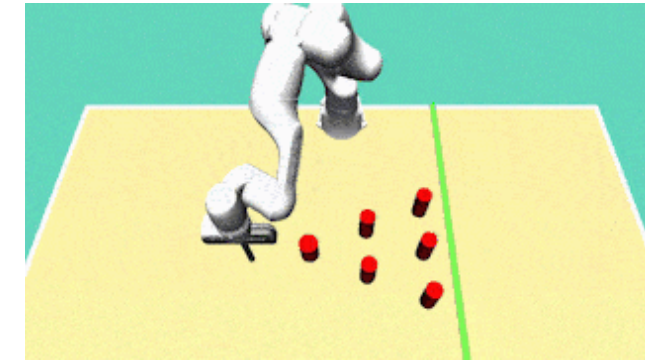
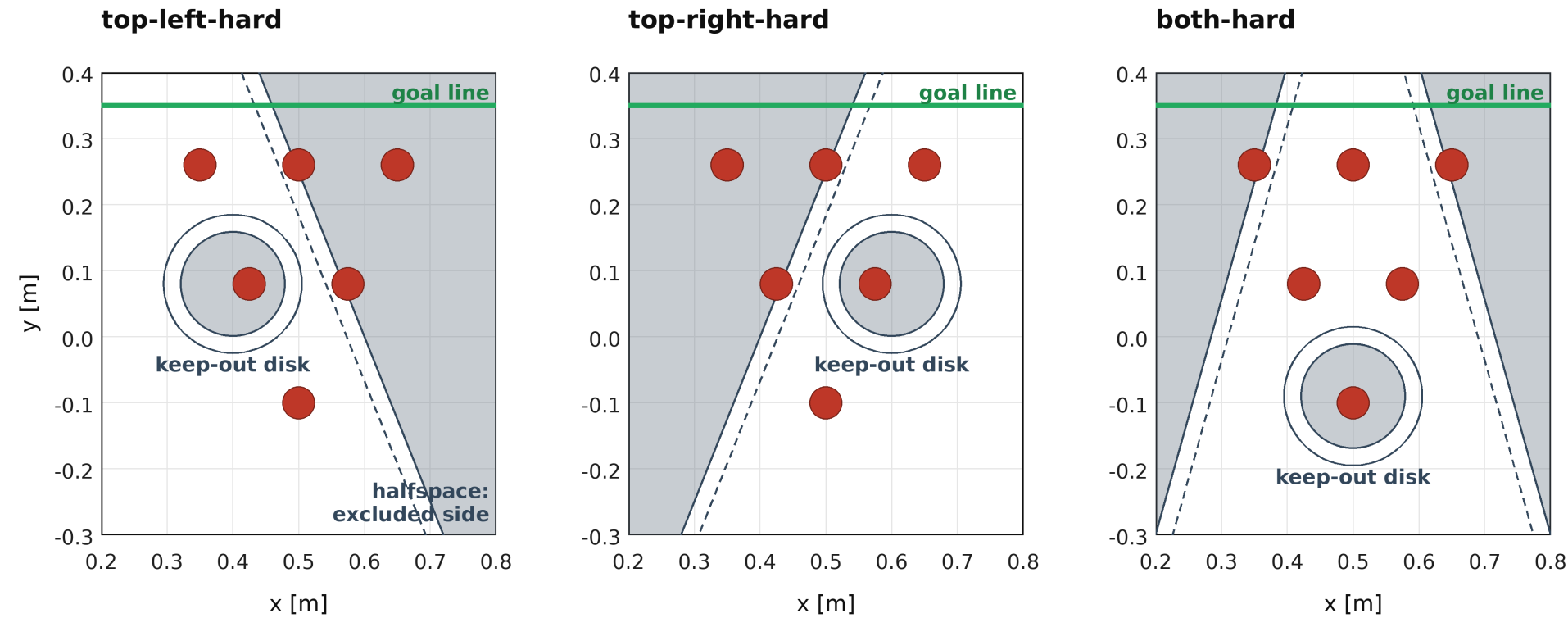
D3IL-avoiding: the benchmark of DPCC, unchanged

■ excluded for the planned position

-- tightened by 0.025 m

● obstacle, radius 0.025 m

— goal line



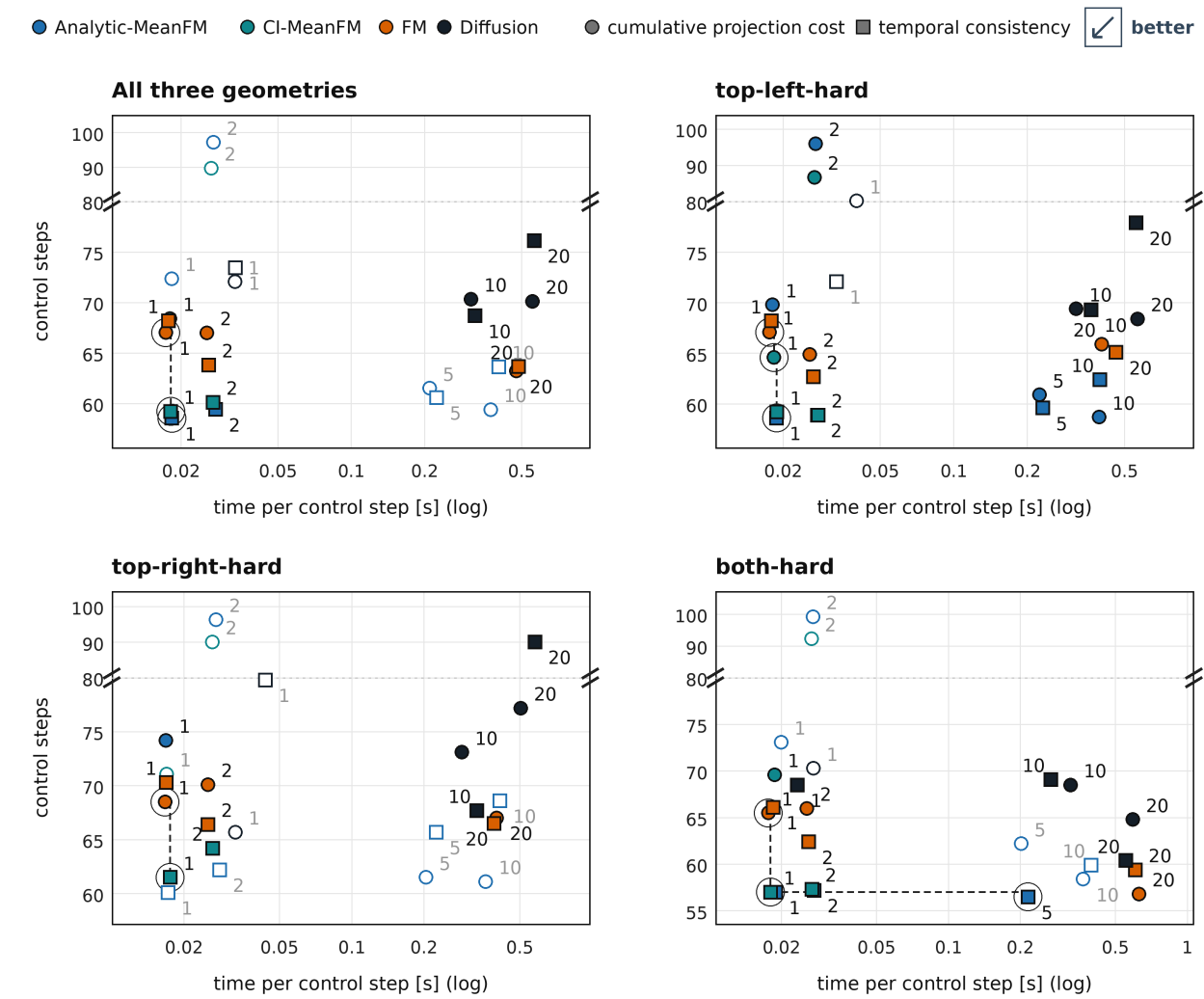
Start pose, goal passed (thesis renders).

MuJoCo rollout GIF of a thesis configuration: pending, to be rendered

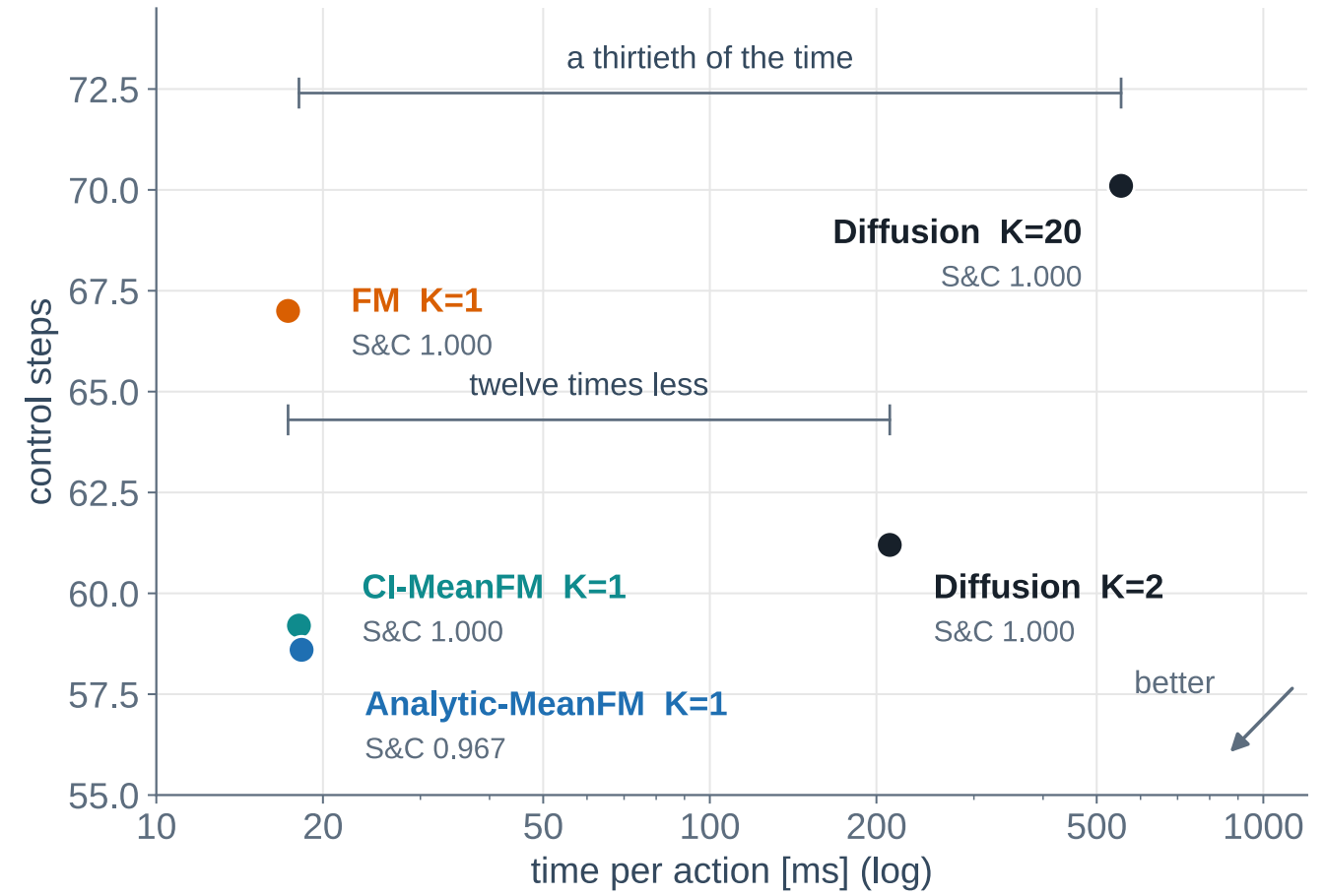
on the cluster.

- The Panda manipulator moves its end effector across a field of six obstacles to a goal line.
- Three constraint geometries; the same environment, demonstrations, projection and protocol as DPCC: five training seeds $\times$ two episodes per geometry.

On the benchmark of DPCC, the average-velocity models dominate its diffusion model



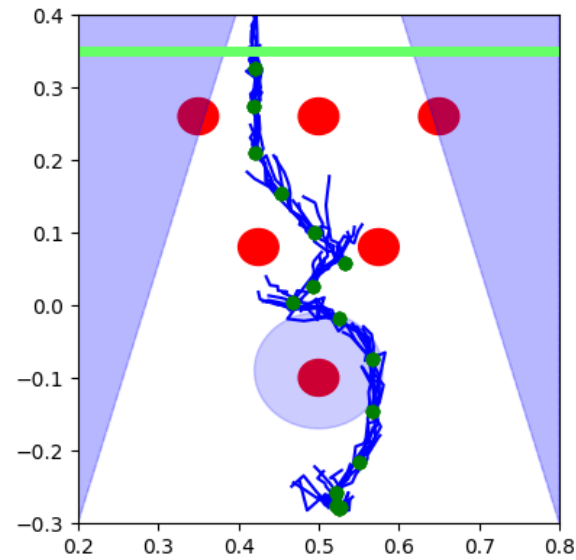
Every budget and rule (thesis figure).



The operating points. Five training seeds $\times$ two episodes per geometry, tightened constraints, temporal U-Net of 4.0 M parameters.

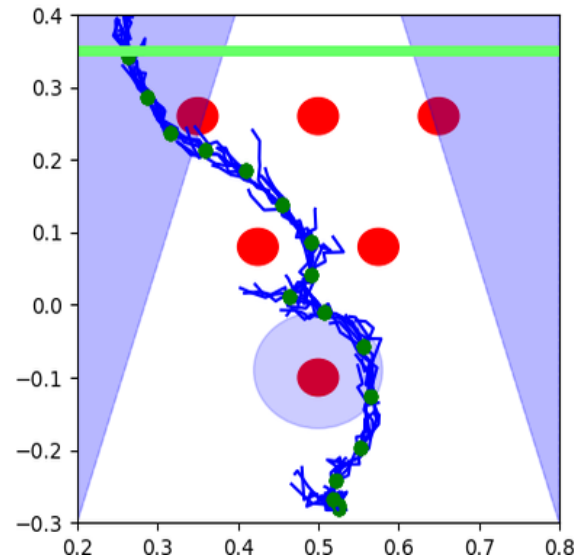
The time a projection takes depends on how far from feasible the plan it receives is

Analytic-MeanFM · $K = 1$ **CI-MeanFM** · $K = 1$

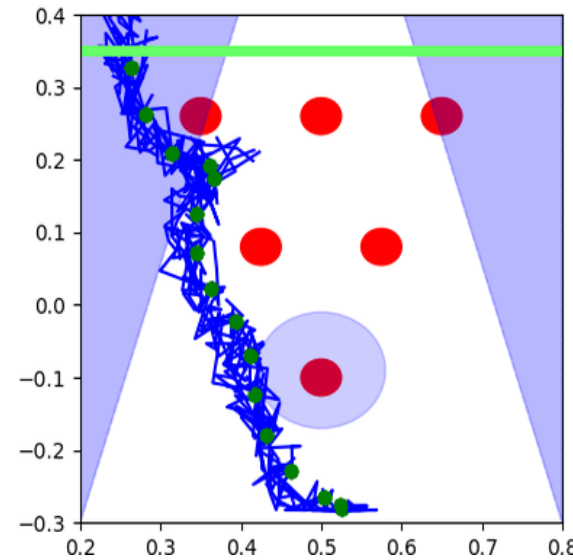


6.0 violating steps

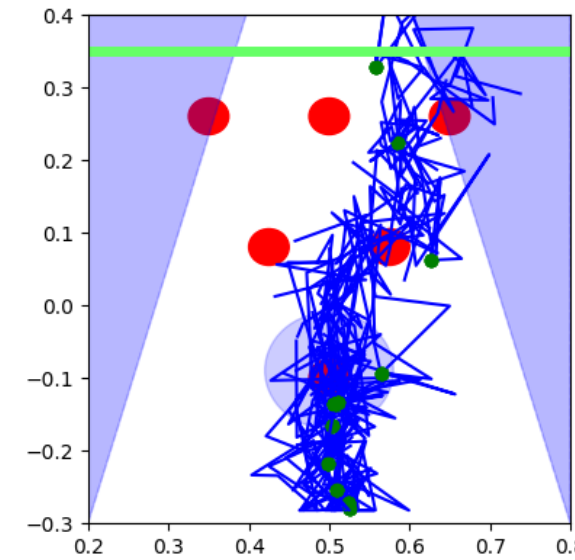
CI-MeanFM · $K = 1$



FM · $K = 1$

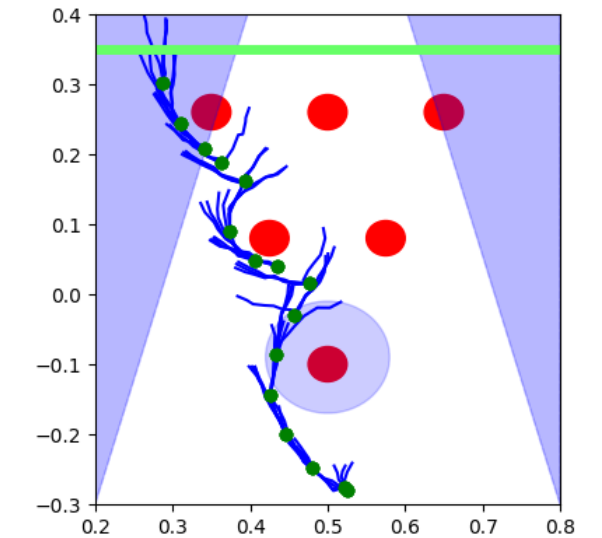


Diffusion · $K = 1$



28.0 violating steps

Diffusion · $K = 20$,
trained budget

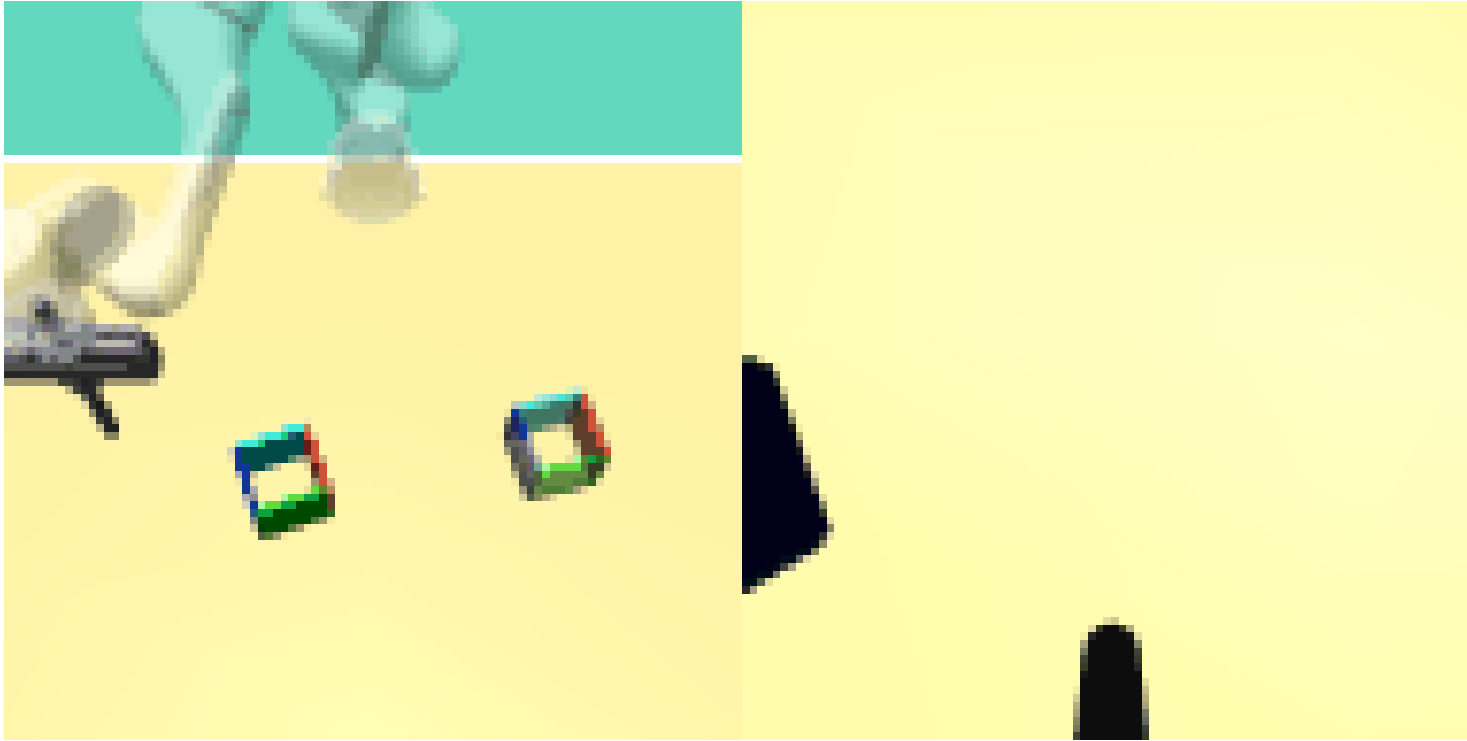


18.5 violating steps

Plans without projection, every fourth control step, all four candidates; both-hard, one training seed; violating steps per episode in this cell. The executed path is not drawn.

- At one evaluation Analytic-MeanFM's successive plans form a narrow ribbon; the diffusion baseline's plans jump across the workspace from one control step to the next.
- At its own twenty steps the baseline draws a single tight bundle, the upper half of which lies inside the left halfspace.

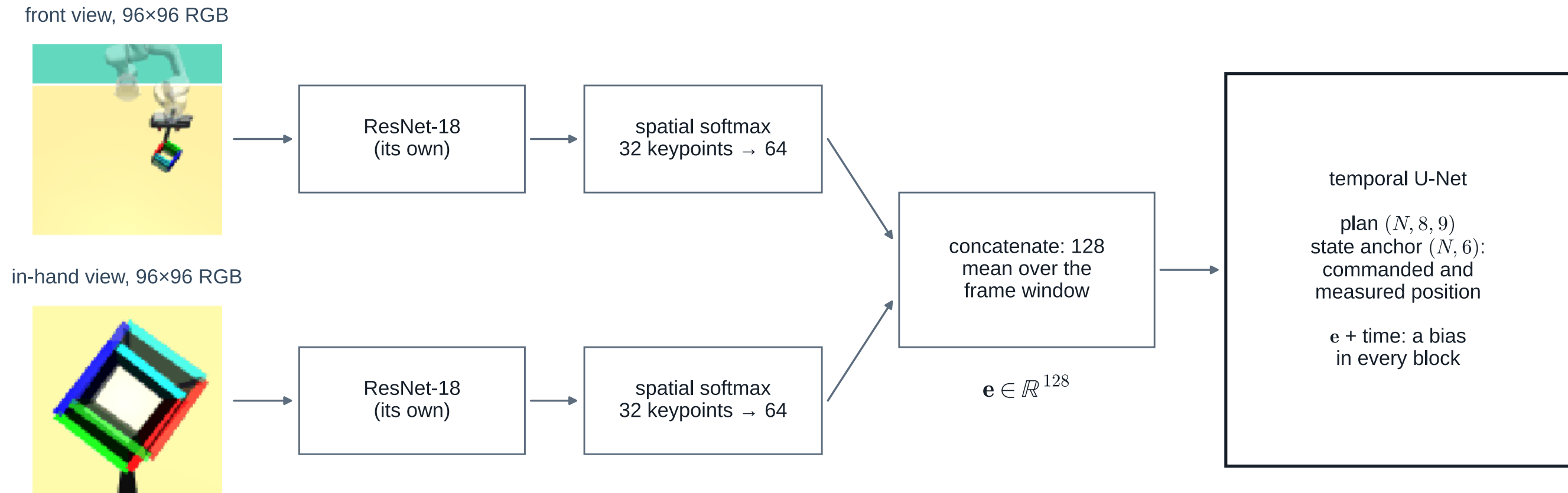
D3IL-aligning: the robot observes two camera views



An expert demonstration of D3IL-aligning in MuJoCo, seen through the two cameras of the observation.

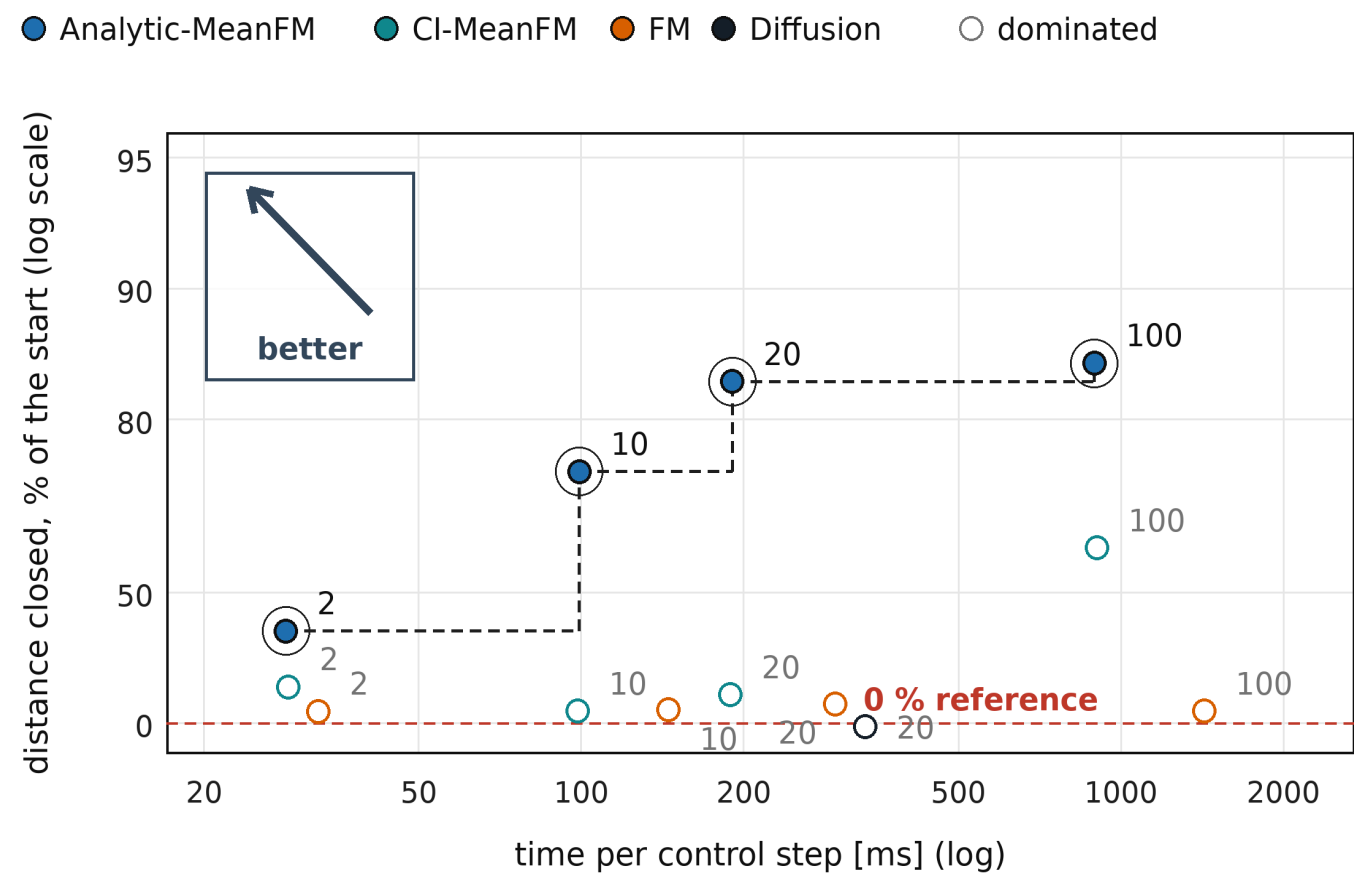
- The manipulator pushes a box into a target pose; each episode starts from a context, an initial box pose and a target pose.
- The observation carries no box or target pose: the policy must locate the box in two RGB streams, a front view and an in-hand view.
- The generative network is given a visual encoder; the projection is given the dynamics rows and the constraint set of the new task.

D3IL-aligning: the images enter the network as one latent of 128 numbers

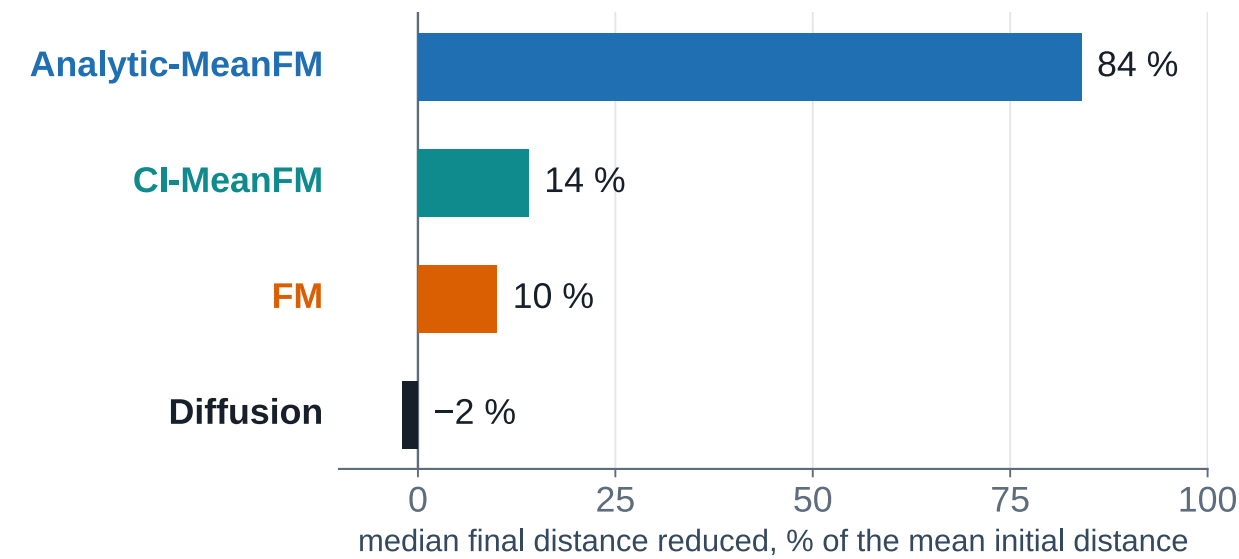


- The observation is two RGB views plus a 6-D state, the commanded and measured end-effector positions; the box and its target pose are only in the images.
- The backbone never sees an image: the latent $\mathbf{e}$ is the interface between perception and generation, and the encoders are trained end to end with the generative model.

On D3IL-aligning the dominance repeats where the task separates the models



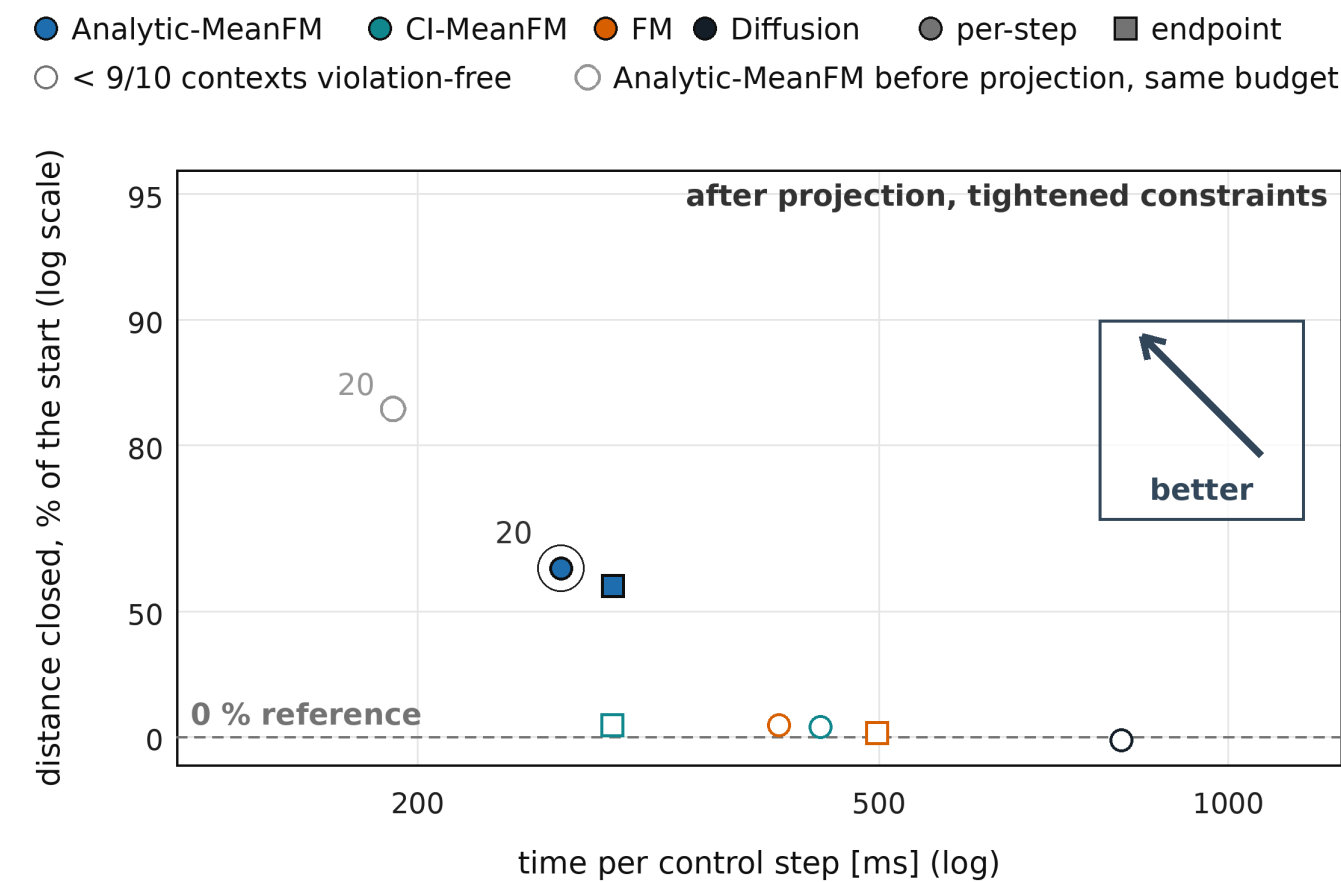
Distance closed against time per control step, every budget (thesis figure).



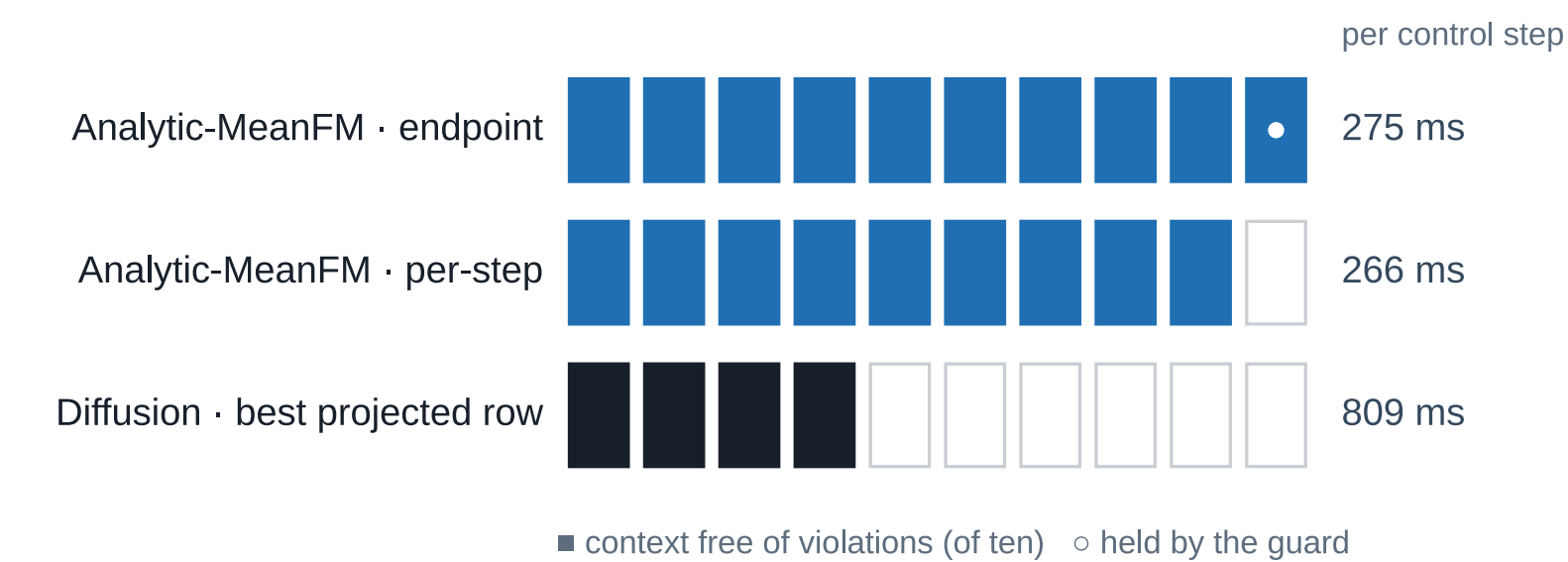
Without projection at $K = 20$, ten training-set contexts.

- At two evaluations, at a seventh to a tenth of their cost, Analytic-MeanFM's reduction is already two fifths.
- **CI-MeanFM** does not improve with the budget below a hundred evaluations.

At the budget this task needs, endpoint projection is the better projector on the constraint



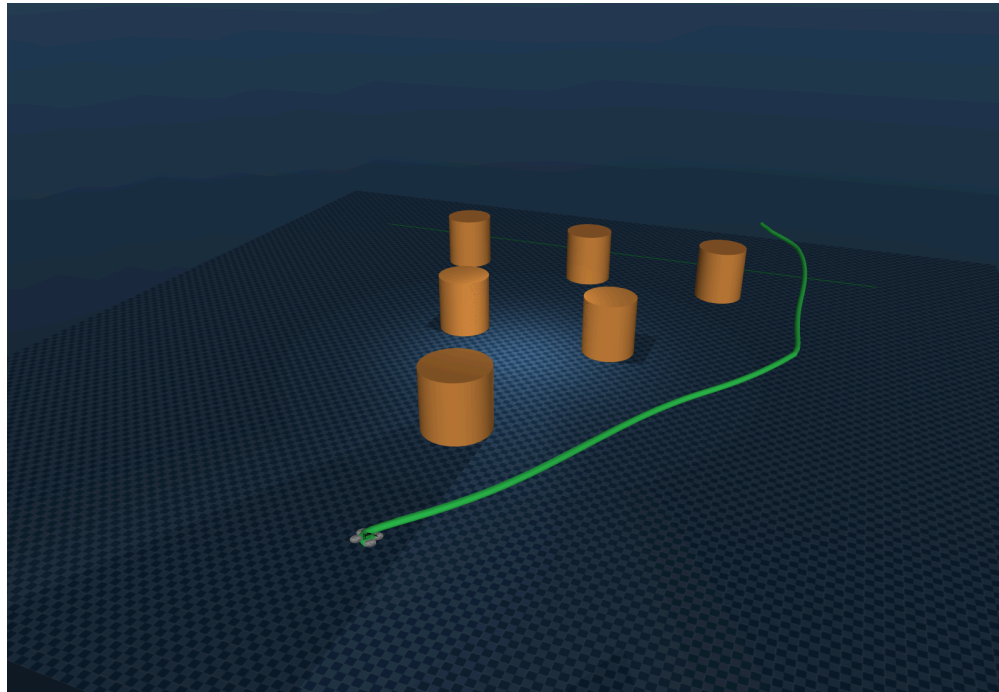
After projection, tightened constraints; circles per-step, squares endpoint (thesis figure).



Contexts free of violations after projection, $K = 20$, random selection.

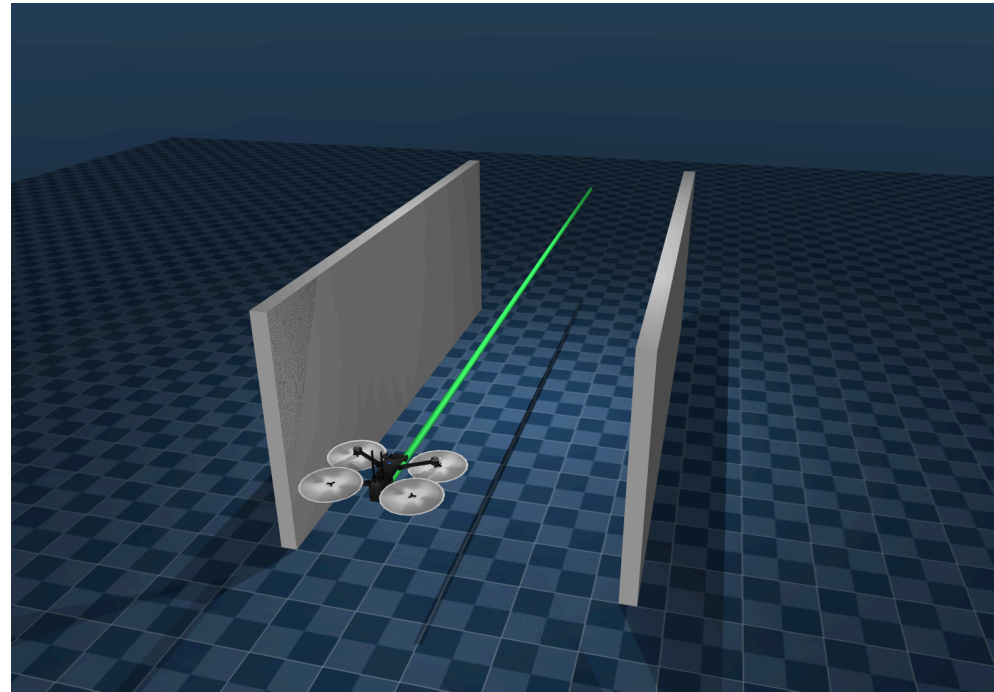
- Endpoint projection keeps more contexts violation-free than per-step projection in seven of nine comparisons.
- The constraint is paid for in distance: no context ends within the 1.8 cm of the task's own test.

Quadrotor: three scenes that ask three different questions



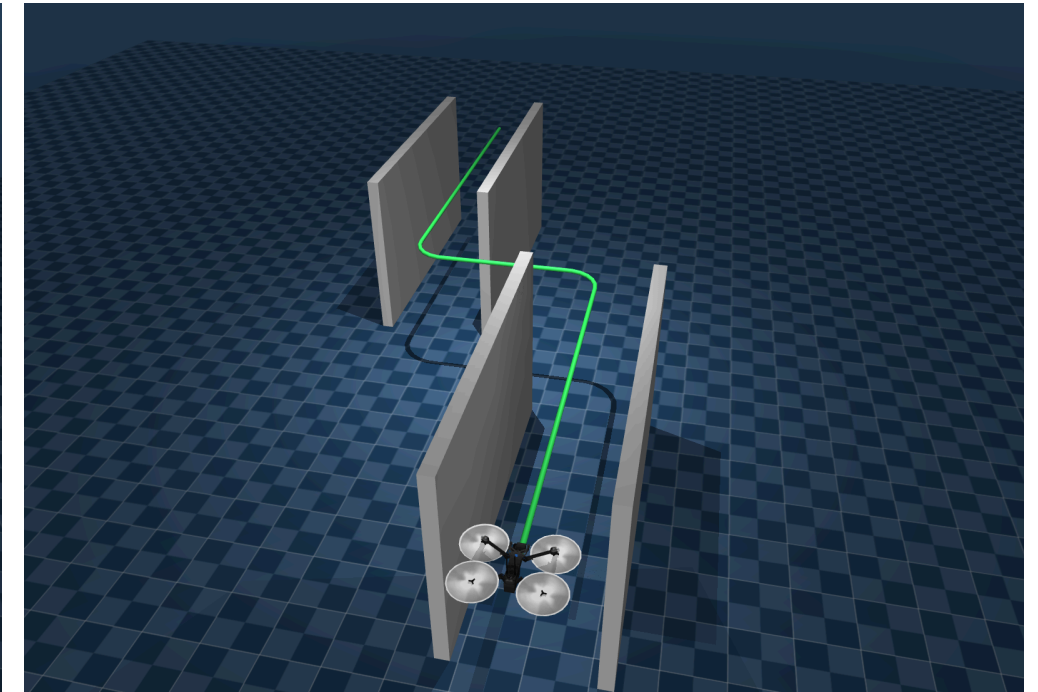
UAV-pillars

D3IL-avoiding flown by the quadrotor



UAV-corridor

constraints that leave the horizontal plane



UAV-s-curve

two sharp turns for the tracking controller

Skydio X2 of MuJoCo Menagerie, commanded through its rotor thrusts by a cascaded geometric tracking controller.

MuJoCo flight demos pending.

The quadrotor keeps the planning side and changes the control problem

D3IL · manipulator



UAV · quadrotor



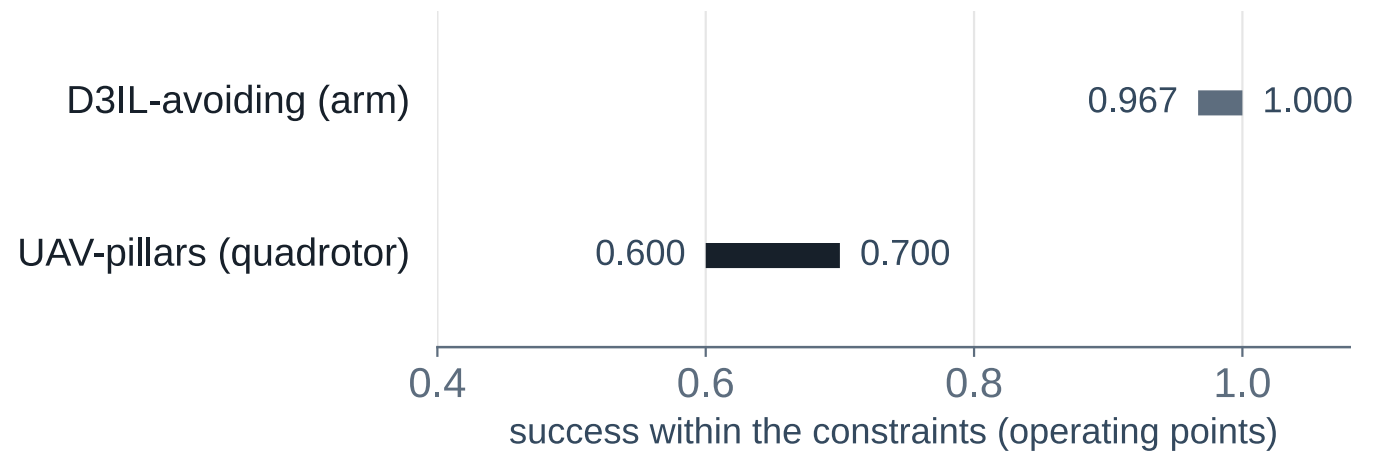
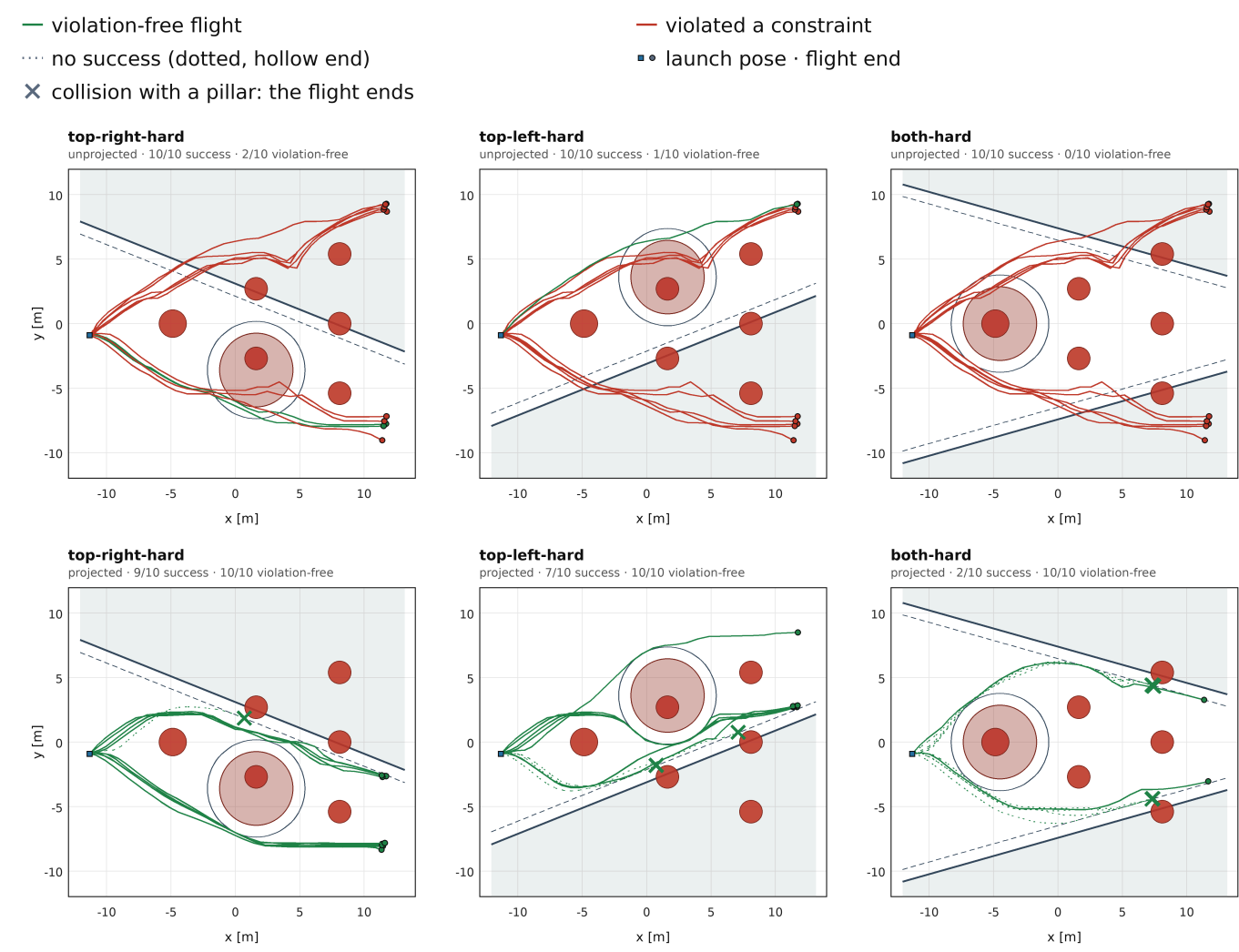
Stays: the planning side

- plans of eight position increments; the same setpoint recursion
- the projection with its first-order dynamics rows ($t_s = 1$)
- the four objectives and their training targets; UAV-pillars flies the D3IL-avoiding models unchanged

Changes: the control problem

- an underactuated, open-loop unstable plant: four rotor thrusts for six degrees of freedom
- the tracking controller: cascaded geometric (MuJoCo MPC in the controller comparison)
- constraints that leave the horizontal plane (UAV-corridor); generated demonstrations

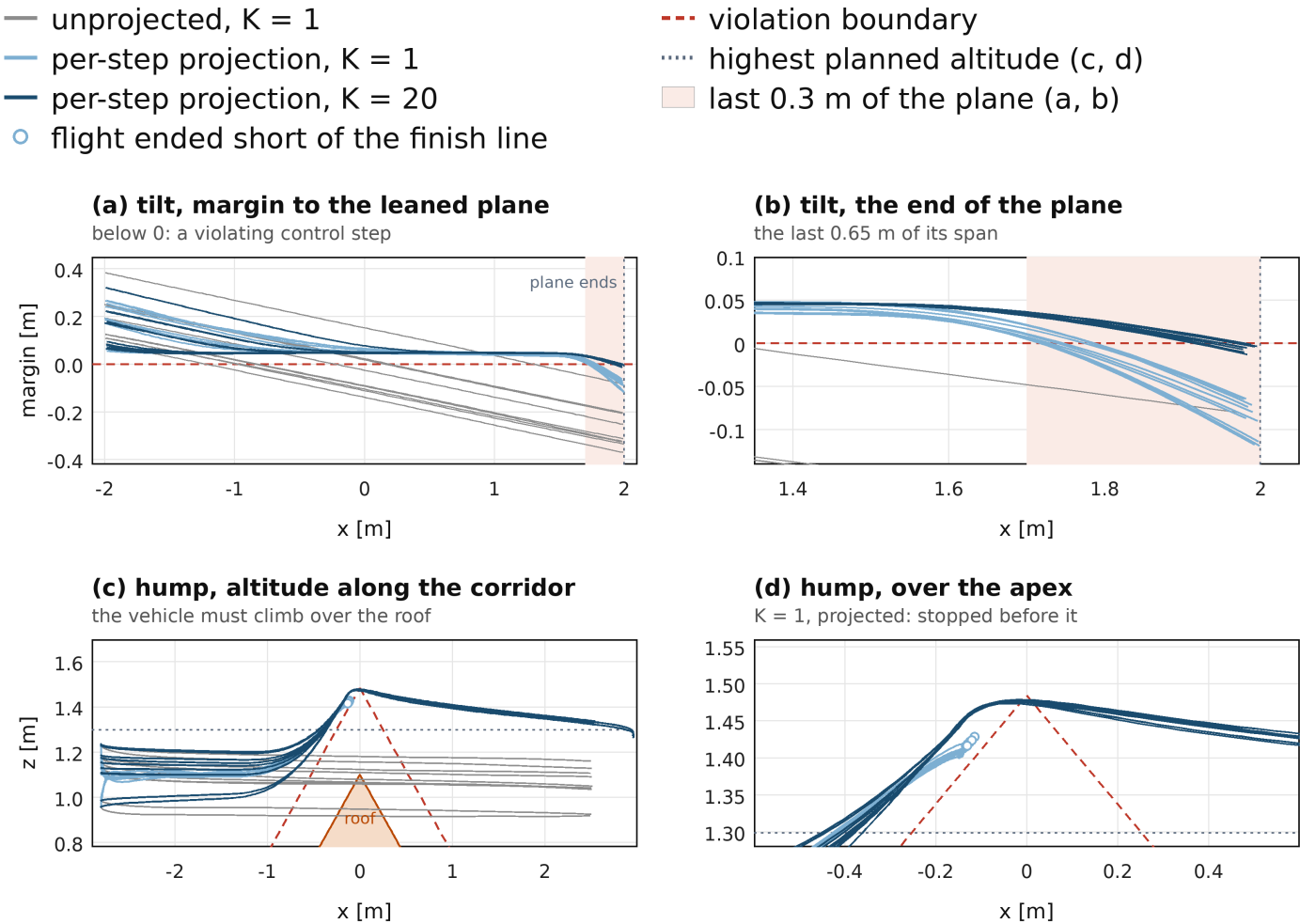
UAV-pillars: the result of D3IL-avoiding holds when its plans are flown



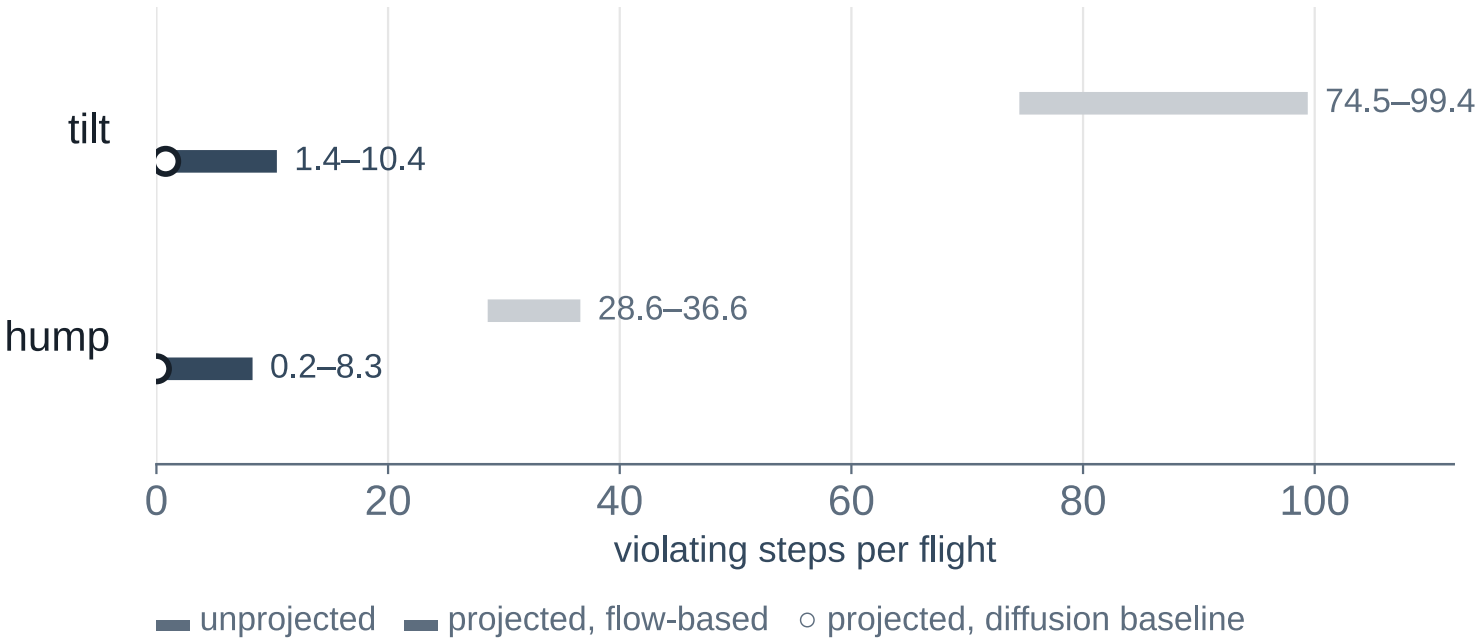
- Success within the constraints at the operating points.
- After projection no flight violates the declared constraints; the vehicle tracks more closely than the arm.
 - Every failure is a collision with one of the five pillars the constraint sets leave out.

Flown paths, unprojected (top) and projected (bottom), three geometries (thesis figure).

UAV-corridor: the projection scheme of DPCC carries over to three dimensions



Margin and altitude along the corridor: unprojected, per-step projection at $K = 1$ and $K = 20$ (thesis figure).

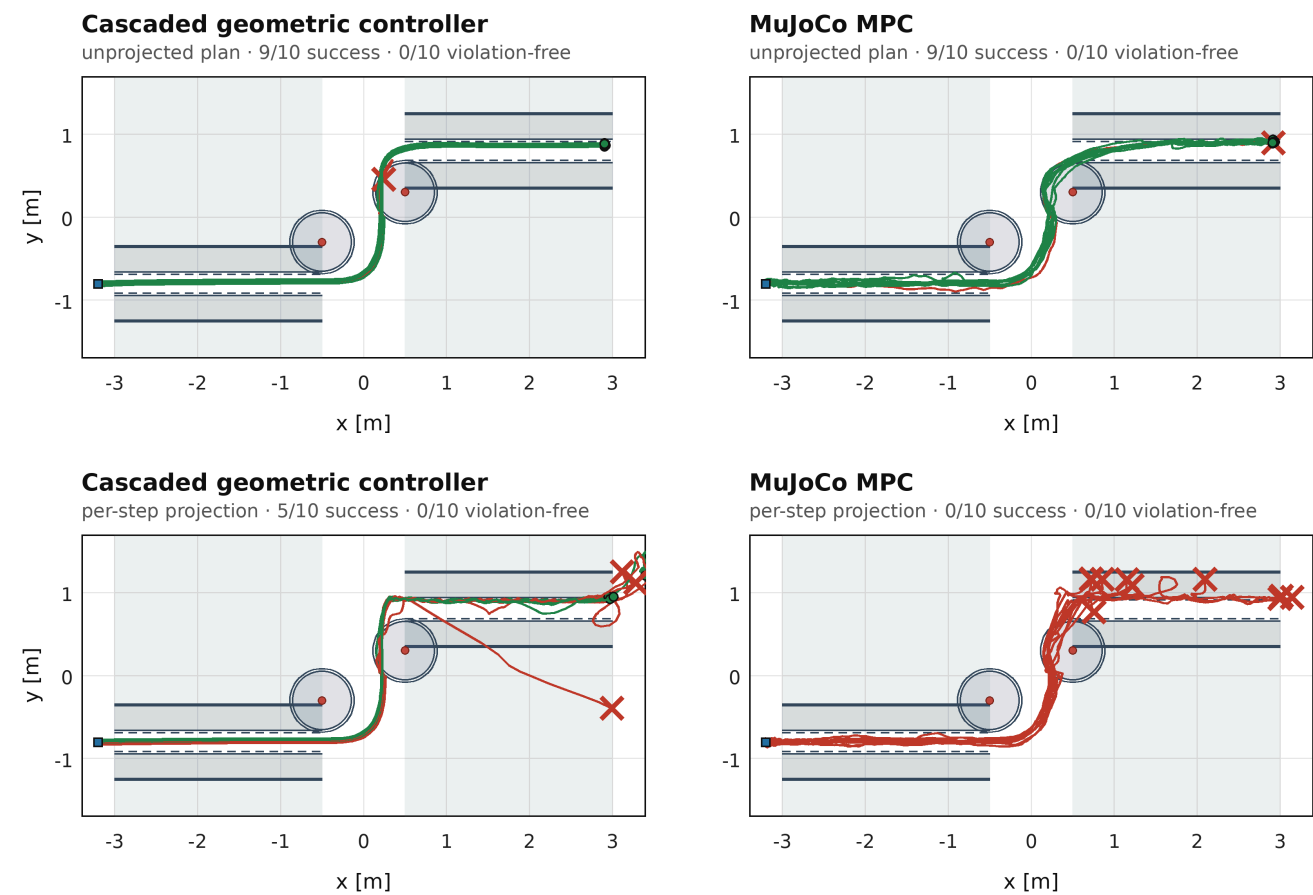


Violating steps per flight.

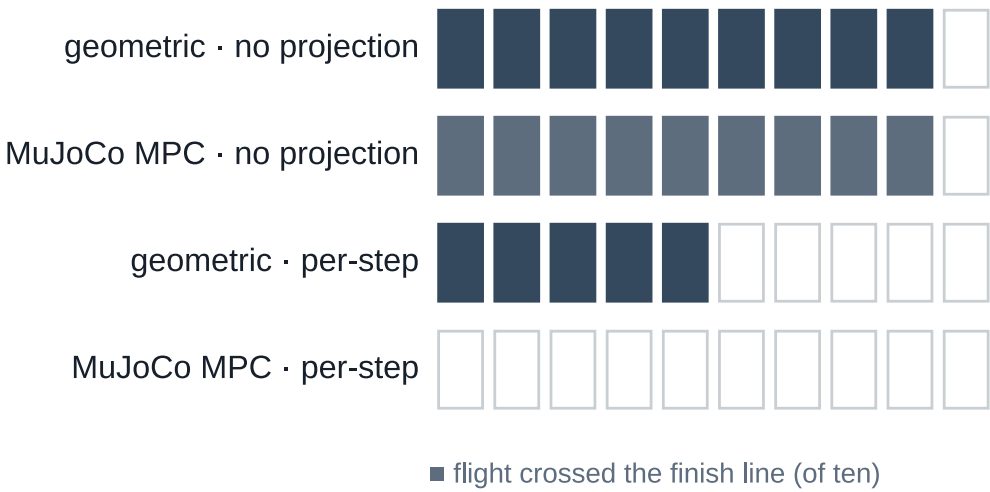
- Projected, the flights descend under the tilt (540 of 560) and climb over the hump (all 560).
- The objectives cannot be told apart, the budget can; between the two projectors a trade-off.

UAV-s-curve: the controller decides whether a plan is flown

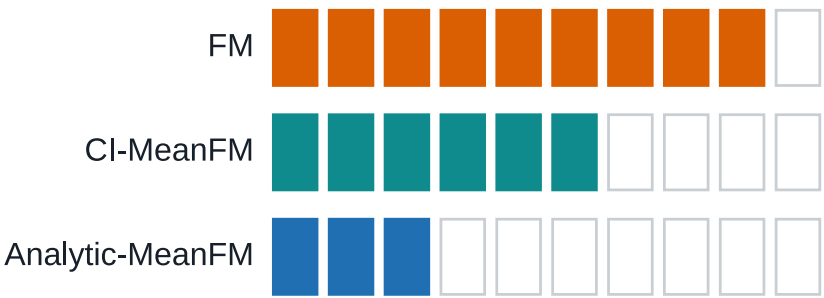
- success: crossed the finish line
- launch pose · flight end
- no success
- × collided or lost control



Cascaded geometric controller (left) and MuJoCo MPC (right); unprojected (top) and per-step projection (bottom) (thesis figure).



The configuration of record, by controller and projection.



at one evaluation, no projection, cascaded geometric controller

By objective.

Three findings across the environments

1. The time a projection takes depends on how far from feasible the plan it receives is.
2. A task saturated at one evaluation hides what a task that rewards more evaluations exposes.
3. The demonstrations bound what a model can show.

Answers to the research questions

- **RQ1 — Yes, on the benchmark of DPCC.** With human demonstrations the average-velocity models come first; with the generated quadrotor demonstrations the objectives tie, or the simplest leads.
- **RQ2 — Where it has steps to guide.** Endpoint projection guides the sample only where the budget and the threshold leave a step before the terminal one; there it is the better projector on the constraint at a comparable planning time. It does not shorten the task, and on the corridor it trades.
- **RQ3 — Where the demonstrations are of the same kind.** The answers hold on camera observations and contact, and the constraints transfer to the quadrotor; the plant and the demonstrations change the rest.

Limitations and outlook

Limitations

- The plan is a sequence of positions; below it is the plant's controller.
- A constraint is what is declared: the projector protects nothing else.
- Every environment is a simulation.

Outlook

- Human demonstrations for the quadrotor scenes.
- A longer planning horizon than eight steps.
- Constraints from perception: depth or range sensing that emits the projector's primitives.

Thank you

Questions